OR Spectrum (2008) 30:1–52
DOI 10.1007/s00291-007-0100-9

REGULAR ARTICLE

# Operations research at container terminals: a literature update

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Published online: 10 October 2007


**Abstract** The current decade sees a considerable growth in worldwide container transportation and with it an indispensable need for optimization. Also the interest in and availability of academic literatures as well as case reports are almost exploding. With this paper an earlier survey which proved to be of utmost importance for the community is updated and extended to provide the current state of the art in container terminal operations and operations research.

**Keywords** Container terminal · Logistics · Planning · Optimization · Simulation

## 1 Introduction

Today, ports are gearing up to meet the challenge of handling mega-vessels capable of carrying 10,000–12,000 TEU[1] and beyond (see, e.g., Baird (2006) for a short overview of increasing ship sizes and traffic growth). Key factors for a container terminal are the efficiency of the stacking and the transport of this large number of containers to and from the ship's side. High productivity and container throughput from quayside to landside and vice versa at low costs are vital for a terminal operator in order

[1] TEU: twenty-foot equivalent unit; a standard size of a container, typically used for denoting the output or capacity of container terminals as well as for defining the container carrying capacity or loading of vessels.

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to compete with other terminals. Furthermore, shipping companies ask for reliability regarding adherence to delivery dates and promised handling times (see, e.g., the eight key determinants of port competitiveness discussed by Tongzon and Heng (2005)). Wang and Cullinane (2006) diagnose that suppliers of container handling services in ports have a weak bargaining position compared to that of shipping lines. Thus, container ports are forced to provide efficient and cost-effective services. They have to invest heavily to meet the stringent demands for faster service and higher quality. Recently, Shi and Voß (2007) discuss some relationships between shipping alliances and container terminal operations.

The goal of this paper is to provide an expository update of research on operations research methods applied on maritime container terminal operations. The paper is an extension and update of Steenken et al. (2004) providing a comprehensive survey of the state of the art of operations at a container terminal as well as of methods for their optimization. We foremost focus on the recent work published after appearance of Steenken et al. (2004). The main benefit of the paper lies in the comprehensive collection of recent references and their classification with respect to the well-accepted structure of Steenken et al. (2004).

Regarding terminal operations and equipment only very few novel aspects such as twin/tandem lift quay cranes are specified in order to complement Steenken et al. (2004). Since we focus on operations at seaport container terminals, ship routing problems or a liner operator's problem of distributing or reusing empty containers, i.e., finding best multimodal container itineraries, are excluded from the survey as well as topics regarding traffic problems in the hinterland, logistics tracking systems, transfer between trains, and similar topics. We refer to, e.g., Chang et al. (2006); Chen et al. (2007b); Choong et al. (2002); Coslovich et al. (2006); Evers and De Feijter (2004); Fagerholt (2004a,b); Imai et al. (2007a); James and Gurol (2006); Jula et al. (2005, 2006); Li et al. (2004, 2007); Martínez et al. (2004); Mattfeld and Orth (2006); Namboothiri and Erere (2006); van Rensburg et al. (2005); Shintani et al. (2007); Song (2006) and Tsai (2006). For a survey of container shipping and port development as well as for aspects of competition and efficiency see, e.g., Ashar (2006); Cullinane et al. (2005, 2006a); Cullinane and Wang (2006); Notteboom (2006a,b); Notteboom and Winkelmans (2001); Notteboom (2004); Notteboom and Rodrigue (2005); Polo and Díaz (2006); Tongzon and Heng (2005); Wang and Cullinane (2006); Yap and Lam (2004, 2006); Yeo and Song (2006). A review of opportunities for the application of operations research methods in intermodal freight transport can be found, e.g., in Macharis and Bontekoning (2004). For an overview regarding status and perspectives of ship routing and scheduling as well as intermodal transportation the reader is referred to, e.g., Christiansen et al. (2004, 2007); Christiansen and Nygreen (2005); Crainic and Kim (2007); Fagerholt and Christiansen (2000); Gunnarsson et al. (2006) and Hsu and Hsieh (2007). For a domain description of the container line industry see, e.g., Bjørner (2007). A comprehensive survey with a focus on the relationship between vehicle routing problems (VRPs) and container terminal operations can be found in Stahlbock and Voß (2007). To make this paper self-contained, some of those findings are included in this paper. For detailed information about worldwide maritime transport trends see, e.g., the current UNCTAD Review of Maritime Transport (via http://www.unctad.org; e.g., UNCTAD (2004, 2005, 2006)) or a study jointly conducted by

Berenberg Bank and HWWI Hamburg Institute of International Economics (Harald et al. 2006) as well as the current annual report of the European Sea Ports Organisation (ESPO) with focus on Europe (ESPO 2007).

This paper is organized as follows. In Sect. 2 we discuss the literature with regard to container terminal systems and their constituent parts such as handling equipment, human resources and assisting systems. In Sect. 3 we discuss research on optimization methods with a focus on particular processes, problems or subsystems at the waterside or landside such as berth allocation, storage and stacking logistics, or horizontal transport. The structure of this section follows the organization and exposition in Steenken et al. (2004) in order to facilitate an understanding of the processes as well as earlier references mainly from 2004 and before by consulting that paper. Furthermore, we discuss research considering integrative approaches such as analytical ones, simulation based ones, and multi-agent systems. In Sect. 4 the paper is concluded with a summary and outlook identifying interesting and promising topics for future research.

## 2 Container terminal systems

The competition between container terminals has increased due to large growth rates on major seaborne container routes. Terminals are faced with more and more containers to be handled in short time at low cost. Therefore, they are forced to enlarge handling capacities and strive to achieve gains in productivity. Different concepts for meeting the current and future demand are utilized. A rather revolutionary approach is the design of new terminals with advanced layouts, e.g., with indented berths, requiring a new infrastructure as well as the automatization in particular in regions with high labor costs. Another approach is the replacement of older equipment with more efficient one. Furthermore, existing infrastructure and equipment can be used more efficiently, e.g., by means of powerful information technology and logistics control software systems including optimization methods (Günther and Kim 2006). Moreover, infrastructure needs to be developed and extended to cope with future developments such as, e.g., the extension of the Panama Canal and the availability of larger and larger container vessels.

### 2.1 Handling equipment

General information as well as particular details of technical equipment for container handling are provided by engineering oriented journals as well as by specialized outlets, brochures, or websites of suppliers of material handling equipment and services in the container sector (see, e.g., http://www.porttechnology.org, http://www.kalmarind.com, http://www.noellmobilesystems.com, http://www.zpmc.com, http://www.cranestoday.com, or http://www.dakosy.de/en). Common equipment such as the chassis-based transporter, straddle carrier (SC), rubber tired gantry crane (RTG), rail mounted gantry crane (RMG; also called automated stacking crane (ASC)) or reach stacker are compared in terms of the actual stacking capacity in Fig. 1.

Modern concepts for container storage and transportation in particular to and from terminals in order to avoid congestion and solve capacity problems of terminals, of

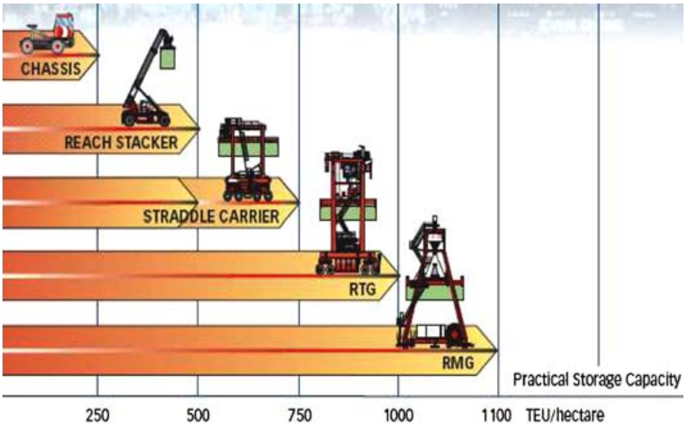


**Fig. 1** Different types of handling equipment and their stacking capacity (Kalmar Industries 2007b)

rails as well as of highways, such as the automated "Freight Shuttle", "CargoMover", "CargoRail", "Auto-GO", "TransRapid for cargo" (maglev freight container movement), "Grid RAIL", Automated Storage/Retrieval Systems (AS/RS) and others, can be found, e.g., in Asef-Vaziri and Khoshnevis (2006); Chen et al. (2003); Dimitrijevic and Spasovic (2006); Hansen (2004); Hu et al. (2005); James and Gurol (2006); Liu et al. (2002); Rohter et al. (2006); Roop (2006) and Zhang et al. (2006). The emerging issues of security with regard to maritime ports and in particular to containers are discussed, e.g., in Banomyong (2005); Bichou et al. (2007) and Tsai (2006). Banomyong (2005) analyzes the impact of the US Container Security Initiative on maritime supply chain management. Tsai (2006) proposes a systematic decision process for constructing a logistics tracking system with automatic vehicle location for preventing the smuggling risk of transit containers in a port.

Hu et al. (2005) analyze the travel time of a new AS/RS. This split-platform AS/RS (SP-AS/RS) is evaluated for high-throughput handling of sea containers. While a conventional AS/RS typically uses stacker cranes for reaching and accessing a storage cell the SP-AS/RS uses separate platforms for high-speed vertical and horizontal movement of heavy loads such as containers. The performance of conventional AS/RS is compared with that of the SP-AS/RS under different rack configurations. Experiments with a sequence of 100,000 jobs show that the new SP-AS/RS mechanism substantially outperforms the AS/RS in terms of throughput and travel time. Furthermore, an SP-AS/RS provides a higher lifting capacity as well as a more flexible rack configuration and high fault tolerance.

Chu and Huang (2005) present a comparison of different container handling systems with regard to a terminal's capacity. The approach aims at supporting decisions on terminal planning with regard to the design of a terminal and the employed handling equipment. Different area requirements for different systems, such as SC systems, RTG systems, and RMG systems are considered. The proposed general equation takes the dimensional characteristics of equipment, the transshipment ratio, and the average container dwell times into account. Findings of the study are that the annual handling capacities increase in order of the SC, RTG, and RMG system, but some values are close or equal among cranes. It is noted that the maximum capacity can vary by more than 100%. This variation is caused by different average dwell times for specific cranes. Hence, selecting appropriate equipment requires a careful evaluation.

Vis (2006a) presents a simulation study for the evaluation and comparison of different terminal systems with manned SCs and RMGs in terms of costs and performances. The task is to perform a fixed number of storage and retrieval requests. The performance criterion is the (average) total travel time including empty and full travel distances, average hoisting times as well as average reshuffle times. Characteristics of each container, such as its location in the stack, the type of operation, or origin/destination are randomly generated. A sensitivity analysis is performed for obtaining fair results. The results show advantages for RMGs for a width of the stack up to nine containers. It is expected that the SC system outperforms the RMG system with a wider stack.

### 2.2 Human resources

Kim et al. (2004a) focus on the efficient scheduling of operators of handling equipment. Major constraints are restrictions on the minimum workforce assignment to each time slot, the maximum total operating time per operator per shift, the minimum and maximum consecutive operating times for an operator, types of equipment that can be assigned to each operator as well as the available time slots for each operator or piece of equipment. From a practical point of view it is satisfactory to achieve a feasible solution within a reasonable time. The problem is formulated as a constraint satisfaction problem to be solved with commercial software (from ILOG®). Numerical experiments are conducted with real data from Busan (Korea) for the evaluation of the search strategies. The problem instances are solved in reasonable time. The experiments show that static variable-ordering leads to a short and steady computation time whereas different priority rules for variables have no impact on the computation time.

Legato and Monaco (2004) study the manpower planning problem at marine container terminals. Particular properties of this problem are the uncertainty of workforce demand as well as the need for ensuring a time continuous efficiency of the terminal. Therefore, the problem is decomposed into two phases with a long-period planning followed by a daily planning. Mathematical programming models for both problems are proposed. They can be efficiently solved by heuristic procedures and mathematical programming techniques. An integer linear programming (LP) problem is proposed for daily planning. The relaxation of a suitable subset of "complicating constraints" leads to a problem which can be solved as an LP problem. A branch-and-bound algorithm is developed in order to solve real-world instances effectively. For long-term planning a heuristic approach to a set-covering type problem is derived. It is stated that the proposed algorithms for both long-term and short-term planning are the core of the crew scheduling system running at the Gioia Tauro terminal (Italy) with a ground crew of about 500 workers. The models can be tailored to similar port structures, in particular to terminals with a lot of transshipment tasks.

Lim et al. (2004a) derive an $\mathcal{NP}$-hard manpower allocation model with time windows from a real-life problem at the port of Singapore. It is similar to the VRP with time windows but a fixed number of servicemen is needed to meet demand. Additionally, a location may have more than one demand overlapping in time. The objectives are the minimization of the number of servicemen used as well as the minimization of travel distances, travel times, and waiting times. The problem features servicemen who are centrally dispatched in order to meet the demand generated from yard locations. A tabu-embedded simulated annealing (SA) algorithm is developed. A squeaky wheel optimization with local search algorithm for the problem is proposed. The effectiveness of the approach is shown by numerical experiments on different sets of generated data. It is noted that no benchmarks are available; the results are better than current results at the port and close to optimum solutions at least for small data sets.

### 2.3 Assisting systems and models

Bassil et al. (2004) study the multi-transfer container transportation and introduce a system architecture of a workflow-based system for the management of container transportation. The modeling and processing of client requests is supported by means of workflow management concepts and technology. The progress of a client request can be efficiently tracked, monitored, and controlled by the users of the system. The requirements for the optimal management of resources during activity planning are met by defining specific optimization models that are data-independent abstractions of optimization problems. An optimization model assigns resources to activities subject to constraints of a client request. Furthermore, the model respects the information related to the transportation network. The resource allocation problem is modeled as a constraint satisfaction problem. It is solved by constraint programming. Time-consuming manual interactions can be reduced by automation of some steps for the administration of the system. Those modification rules are represented and exploited by means of a rule engine.

Kim et al. (2004b) present an architectural design of software for control of automated container terminals. Moreover, an object-oriented simulation-based testbed for the evaluation of various control rules is introduced. The structure of a terminal's control software is quite versatile due to the large number of different types of the terminal's handling equipment. The proposed architecture is subdivided into a ship operation manager for dispatching tasks to equipment for pre-planned (un)loading operations, a system controller for automated guided vehicles (AGVs), and a controller for automated yard cranes. Equipment is assigned to the ship operation manager by both system controllers that keep track of the location and status of the equipment. The simulation environment allows testing of operation-related issues and strategies for controlling traffic at the terminal. Three traffic control rules are introduced with promising results: the synchronization of operations and events regarding quay cranes, AGVs and yard cranes, the re-sequencing of AGVs entering transfer areas under quay cranes with buffers, and the postponement strategy.

Ngai et al. (2007) discuss a case study on the development of a prototype system in a container depot that uses radio frequency identification (RFID) features. Benefits and advantages as well as problems of the RFID-based approach are discussed. For example, the system supports tracking of the locations of stackers and containers.

Furthermore, it improves the visibility of operations data as well as of control processes. In particular, mobile commerce activities in a container depot are supported by the system. Disadvantages of typical information systems regarding storage data, such as misplaced containers or inefficient search for containers, are avoided in processing order requests.

Cho et al. (2007) propose a framework for analyzing container terminal performance. The approach aims at figuring out best practices and providing benchmarks for decision making. The integrated analysis and simulation methods are based upon the Delphi hierarchy process method with a dynamic environment. The method combines the well-known Delphi method using an expert panel for data collection, i.e., compiling and weighing performance criteria, and the mathematical model of the (fuzzy) analytic hierarchy process for the evaluation and integration of results. Example results are shown for Busan port container terminal cases.

Hwang et al. (2007) propose a performance evaluation model for container terminal systems. The approach aims at relaxing the common assumption of permanent availability of resources. Therefore, performance factors, such as the system configuration, the system reliability, availability and maintainability (RAM), the life cycle cost, and the system optimization, are considered in a comparative approach with four steps. In the first step, an initial system configuration to meet the required production rate is found by means of a closed queueing network model. This system configuration is simulated in order to improve its robustness by considering the system RAM (step 2) and life cycle cost (step 3). The final system performance analysis is done with a simulation approach. Results of numerical experiments show that the best system configuration can be found as well as maintenance policies for meeting the required process rate.

## 3 Optimization methods for terminal logistics

### 3.1 The ship planning process

#### *3.1.1 Berth allocation*

Dai et al. (2004) formulate the static berth allocation problem as a rectangle packing problem with release time constraints. The problem is solved using a local search algorithm. The neighborhood structure is defined by augmenting the sequence pair approach proposed by Imahori et al. (2003) based upon the approach of Murata et al. (1996) representing a solution by a pair of permutations of rectangles. It is shown that a berthing plan can be encoded by a pair of permutations of all vessels. Furthermore, the approach is embedded in a real-time scheduling system to reflect dynamic aspects and allow for determining the vessel allocation and the planning horizon as well as for choosing between the objective of minimizing a ship's waiting time or maximizing the berth utilization. Results from simulation experiments are conducted for a moderate load and a heavy load scenario. In the former scenario berth space can be allocated to most of the calling vessels upon arrival (>90%). Most of the assigned locations (>80%) are preferred ones. In the latter scenario the concerns of throughput have to be balanced with the acceptable waiting time experienced by vessels. It is shown that deliberately delayed berthing of vessels is an appropriate way to achieve higher throughput. It is concluded that the approach is limited for a real-world application since important berthing constraints, such as the crane availability or the separation distances between berthing ships, are ignored. Future research will tackle this problem as well as the problem of dealing with erratic port stay times.

Henesey et al. (2004) propose a berth allocation management system for simulating and evaluating berth allocation policies at a container terminal. Different scenarios with various quay lengths, berth spacing lengths, and ship arrival sequences are treated with two berth assignment policies ("shortest turn-around time" and "berth closest to stack"). The system helps in analyzing the impact of assignment decisions on resources, such as berth space and cranes. The policies are evaluated in terms of the vessels' turnaround time and distance traveled by the SCs. The results of the simulation experiments indicate that an informed choice of berth assignment policy can reduce the turnaround time and improve the utilization of SCs. Thus, a terminal's efficiency can be increased without cost intensive structural changes.

Cordeau et al. (2005) present a tabu search (TS) algorithm for solving the berth allocation problem. The study is based upon data from a terminal in the port of Gioia Tauro. Drawbacks of the model of Park and Kim (2003) for simultaneously solving the berth assignment problem and the quay crane allocation problem are briefly discussed. In contrast to their approach, the authors prefer the development of an efficient heuristic algorithm for the berth assignment problem as a first step. In a subsequent study, a decision support system (DSS) for solving the quay crane allocation problem with consideration of nonlinearities should be devised. Furthermore, the proposed approach for the berth assignment differs from the mixed integer programming (MIP) model and SA approach proposed by Kim and Moon (2003). While in Kim and Moon (2003) and Park and Kim (2003) each berthing point is penalized by a factor proportional to its distance from an ideal point Cordeau et al. increase the expected handling time for a vessel according to the quay segment where the vessel moors. The dynamic berth allocation model of Imai et al. (2001, 2005a) is considered as well as the formulation as a multiple depot VRP with time windows (MDVRPTW). Ships representing customers are modeled as vertices in a multigraph. Service time windows on the ships are expressed by imposing time windows on those vertices. A berth is seen as a depot that is divided into an origin vertex and a destination vertex. Time windows on those vertices correspond to the availability period of a berth. There is one vehicle for each depot and tours start and end at the vehicle's depot. The objective is to minimize the weighted sum of service times. Five problem sizes are considered for the discrete case (25 ships and 5, 7, 10 berths; 35 ships and 7 and 10 berths), and ten instances are randomly generated for each problem size. For the continuous case 30 instances are randomly generated. The TS algorithm always yields optimal solutions on small instances of the discrete case. The truncated CPLEX® algorithm is always outperformed on larger instances. For the continuous case results of the modified TS algorithm are compared with results from the discrete case and from a simple constructive procedure. This comparison differs from the comparison in Kim and Moon (2003) since only very small instances (7 ships, scheduling horizon of 72 hours) of the MIP model are solved exactly. The findings of the experiments are that more realistic instances can be solved by the proposed heuristics compared to methods being previously published by other authors. It is stated that the terminal in Italy plans to incorporate the proposed heuristics in its DSS. Various practical features, such as time windows or favorite and acceptable berthing areas, can be handled by the heuristics. Further research will focus on the integration of the quay crane assignment problem.

Cordeau et al. (2007) discuss the tactical service allocation problem at a container transshipment terminal based on experience at the Gioia Tauro port. The service allocation decisions are usually made over a 3 month horizon. It is the yard management's problem of fulfilling requests of shipping companies for dedicating favorite areas of the yard and the quay to their port route. Decisions on this tactical level may be modified when vessels actually arrive at the port. Thus, on the operational level the berth allocation problem must be solved for avoiding traffic congestions and reduction of service time. The service allocation problem can be formulated as a generalized quadratic assignment problem (GQAP; introduced by Lee and Ma 2005) with side constraints. In the GQAP, $n$ weighted facilities, $m$ capacitated sites, a traffic intensity matrix between facilities, a distance matrix between sites, unit traffic costs, and assignment costs of facilities to sites are given. The objective is the determination of an assignment of facilities to sites while minimizing the sum of assignment and traffic costs. Furthermore, the total weight of all facilities assigned to the same site must not exceed the site capacity. Thus, the GQAP generalizes the quadratic assignment problem in which $n = m$ and exactly one facility must be assigned to each site. In Cordeau et al. (2007) the objective is the minimization of the container rehandling operations inside the yard. In a first mixed integer linear programming formulation (MILP), the berth and the corresponding yard positions extend along a line (like at Gioia Tauro). The second MILP adapts a linearization for the GQAP based on linearizing the quadratic assignment problem. An evolutionary heuristic is developed since optimum solutions can be found only for small instances. It yields optimal solutions for small instances and outperforms the CPLEX® truncated branch-and-bound algorithm for larger instances. It is concluded that the proposed heuristic is appropriate for real-life instances. It fills the gap between the large amount of data provided by terminal information systems and the current practice of applying rules of thumb.

Laganá et al. (2006) focus on the power of grid computing for solving simulation optimization problems of stochastic nature such as the assignment of berth slots and cranes to shipping services. The proposed queueing model is a simplification of the model proposed by Legato and Mazza (Legato et al. 2001). An algorithm for distributing the computational load to parallel processors is developed and tested in computational experiments with problem instances of real-world size.

Moorthy and Teo (2006) propose a framework addressing the home berth template design problem, i.e., the problem of the allocation of the preferred berthing location to a set of vessels scheduled to call at the terminal on a weekly basis. Economical impacts are analyzed. A trade-off between the two dimensions of a terminal's service expressed as waiting time for vessels and operational cost induced by container movements between berth and yard is shown. The problem of stochastic deviations of expected arrival times resulting in last-minute ad-hoc replanning is noted as well, leading to cost-extensive buffers and additional resources, respectively. The approach aims at creating robust berth templates resulting in better service levels as well as in better resource management. The bi-criteria optimization problem is modeled as a rectangular packing problem on a cylinder. It is solved by a sequence-pair-based SA algorithm turning out to be efficient in extensive computational experiments. Results show that the dynamic impact of stochasticity can be effectively measured. Furthermore, overlaps in the template can be minimized. Further research will extend the model for creating templates for vessels with different periods. A more sophisticated model will be necessary for the integration of the crane allocation problem.

Bae et al. (2007) investigate a dynamic berth scheduling method for minimizing the travel costs of vehicles during the ship operation, the tardiness costs, the earliness costs as well as the costs for a vessel's waiting time. Real constraints and various dynamic situations are taken into account. The similarity and difference between the berth scheduling problem with determining locations of vessels in space and time and the space allocation problem, the median location problem, the facility layout problem as well as the spatial scheduling problem are shown.

Imai et al. (2005b) present a heuristic for the continuous berth allocation problem. The paper is the consequent enhancement of previous papers (Imai et al. 1997, 2001, 2003, 2005a; Nishimura et al. 2001) considering discrete locations. Those approaches are easy in scheduling but the terminal usage is not fully efficient. The proposed approach aims at higher flexibility in berth allocation planning that is mandatory in particular for busy hub ports with vessels of various sizes. It is assumed that a ship's handling time depends on its quay location (due to a trailer fleet that is usually limited in size). Furthermore, it is assumed that ship arrivals are dynamic, i.e., the study does not assume any committed ship departure times. Since the research focuses on the total efficiency of ship handling by minimizing the delay in ship departure, the number of violations of committed departure time is implicitly minimized. The developed heuristic incorporates a process of two stages: at first, an algorithm for the discrete quay location finds a solution given the number of partitioned berths. Those ships that may overlap or be located sparsely in a scheduling space are relocated within the second stage. Numerical experiments show that the heuristic works well and meets practical requirements. While this approach deals with conventional forms of terminals and a nonlinear mathematical programming formulation, Imai et al. (2007b) propose an enhanced approach. An integer LP formulation is found for an easier calculation. Furthermore, this formulation is extended to deal with indented berths for fast handling of mega-containerships and feeder ships for achieving higher berth productivity. Numerical experiments with genetic algorithms (GAs) are conducted for indented and conventional terminals of the same size. It is concluded that single mega-vessels can be served faster at indented terminals but the total time for all ships is longer than at conventional terminals. The indented terminal is less efficient for an entire shipping service including mega-vessels and small feeder ships. The indented terminal shows disadvantages in particular in terms of the total service time and transit time of an entire shipping network. According to Ceres Paragon Terminal BV (2006), the world's first indented berth is located at the Ceres Paragon Terminal (Amsterdam). It was planned for doubling the productivity due to the berth's capability of serving vessels from both sides (see Fig. 2). However, we have not yet seen respective results.

Imai et al. (2006a) present a variation of the berth allocation problem as ships are assigned to an external terminal due to berthing restrictions at the usually used berth.

**(a)** Aerial view of the terminal area (Ceres Paragon Terminal BV 2006, p.121)

**(b)** Quay cranes can serve a vessel from (Ceres Paragon Terminal BV 2006, p.123)

**Fig. 2** Ceres Paragon Terminal (Amsterdam) with an indented berth

The total service time of ships at the external terminal is minimized by means of a heuristic based upon GAs with character string representations of chromosomes. In general the total handling time incorporates the ship's waiting time for berth availability as well as the handling time at the allocated berth. Since the waiting time is ignored in Imai et al. (2006a) the total service time equals the total handling time. Numerical experiments with key data of the port of Colombo (Sri Lanka) representing arrival and handling of 56 ships calling for 10 days show good performance of the algorithm as the usage of the external terminal is reduced.

Lokuge and Alahakoon (2004, 2005, 2007) and Lokuge et al. (2004), respectively, focus on automated berth scheduling and monitoring in container ports. They aim at improving decisions on berth assignments in order to increase the berth productivity by investigating the possibilities of using intelligent software agents for managing port operations. The agents are able to autonomously adapt to a changing environment. This approach is motivated by specific characteristics of the complex vessel berthing problem, such as the necessity to deal with uncertainties, and the dynamic behavior of the system. The berthing system is responsible for computing the expected times of berthing and completion, the expected berth productivity, the allocation of a berth as well as the allocation of equipment, such as cranes and vehicles, and human resources to operate them. The implementation of conventional software is expected to be costly and difficult as intelligent human intervention is required for managing the dynamic system's behavior. The common agent architecture of beliefs, desires, and intention (BDI) is limited for complex applications that must learn and adapt their behaviors in making rational decisions. Therefore, a new hybrid BDI framework with an intelligent module is developed. A knowledge acquisition module is introduced, and an artificial neural network as well as an adaptive neuro-fuzzy inference system (ANFIS) are used. A motivation-based distance calculation method supported by ANFIS and reinforcement learning is proposed in Lokuge et al. (2004). It improves the reactive, proactive, and intelligent behaviors of generic BDI agents in complex applications. Results of experiments on a berthing application are discussed. Different scenarios based upon data from Colombo's Jaya terminal (Sri Lanka) are simulated with the proposed agent architecture demonstrating success of the approach.

Wang and Lim (2007) transform the $\mathcal{NP}$-hard berth allocation problem into a multiple stage decision-making procedure. For solving this problem, a stochastic beam search algorithm is discussed in order to cope with the difficulty of finding effective bounds or accurate estimations on node quality. Furthermore, an improved beam search scheme, a two-phase node quality estimation, and a stochastic node selection criterion are proposed. The performance is compared with a state of the art SA from Dai et al. (2004). Results from computational experiments using real data from Singapore show that the stochastic beam search outperforms both the SA and the common deterministic beam search in terms of accuracy and efficiency. Furthermore, it can be easily modified and reused after revisions of the objective function as well as implemented, tested, tracked and tuned.

### *3.1.2 Stowage planning*

Ambrosino et al. (2004) consider the stowage planning problem while taking a set of structural and operational restrictions into account. The problem is denoted as "master bay plan problem (MBPP)". The objective is the minimization of the total stowage time. Practical constraints such as different dimensions or weights of containers are considered. A basic 0/1 LP model is proposed. Since this model is not practically useful for large instances, a heuristic is proposed for preprocessing. Thus, pre-stowage rules are applied allowing to relax some constraints of the exact model resulting in solutions of the combinatorial optimization problem. The approach strives to avoid unloading-related rehandles by assigning some ship holds to containers with the same destination. Loading-related rehandles that occur if specific containers are stacked on the yard below others which are to be picked up later are not considered in the approach. The approach is evaluated with real size cases based upon data from a terminal in Genoa (Italy), showing good performance in terms of precision and computational time. The terminal performance in terms of handling operations per hour can be improved. Future research will evaluate the impact of the ship system requirements on the entire yard organization. In Ambrosino et al. (2006) the authors present a three phase algorithm for defining stowage plans. The objective is the minimization of the total loading time. Weight, size, and stability constraints have to be satisfied related to weight distribution. In the first phase, a main branching tree splits the ship into different partitions. Containers with different subsets of bays are associated without specifying their actual position. Thus, the search space for the following procedure is limited to portions of ships. The successive solution of the 0/1 LP model results in the optimal plan for loading containers into each partition of the ship. Finally, a local search and exchange procedures are performed for checking and removing possible infeasibilities of the global solution caused by stability constraints. The approach facilitates parallel loading operations of different ship portions and shows good performance in numerical experiments based upon data from Genoa.

Sciomachen and Tanfani (2006) utilize the relation between the MBPP and the three-dimensional bin packing problem. Containers are regarded as items, the only bin is the vessel. The approach aims at optimizing important terminal performance measures, such as crane productivity. Structural and operational constraints of containers and ships are taken into account. Thus, this paper is part of the research announced in Ambrosino et al. (2004). Again, the approach is validated with data from Genoa. A comparison with results in Ambrosino et al. (2004) shows effectiveness of the new approach.

Imai et al. (2006b) focus on container stowage and loading plans of a ship. Two conflicting criteria are taken into account: requiring the minimum number of container rehandles (loading-related and unloading-related ones) in yard stacks as well as meeting ship stability. A multi-objective integer programming model is proposed. The weighting method proposed by Cohon (1978) is used to obtain a set of non-inferior solutions. In the weighting method the problem is defined as a mathematical programming model with a single objective incorporating multiple objectives. Numerical experiments show that the solutions are acceptable for practical purpose if no rehandles occur during unloading. Furthermore, the GA is tested with and without a tournament approach showing that the tournament provides better solutions but needs more computation time.

Álvarez (2006) proposes an approach using TS and multiple initial plans for automated generating of vessel loading plans in reach stacker based terminals. For judging the quality of alternative solutions, the algorithm takes the impact of container reshuffling, ground travel of reach stackers as well as on-board weight distribution into account. The algorithm's performance is evaluated in comparison with an MIP formulation of the problem. Computational experiments using data from a Spanish terminal show that the algorithm can provide good solutions for test instances with hundreds of containers within a few minutes. The basic MIP formulation failed to solve problem instances with more than 40 containers. Alternatively, Álvarez (2007) proposes a Lagrangian relaxation approach for the problem. Powerful lower bounds are obtained by the algorithm within a branch-and-bound framework. Subgradient optimization is applied to the relaxation sub-problems. Furthermore, important implementation issues for speeding up the algorithm are discussed. Computational results on small test instances (10, 15, 19 containers) are conducted for a comparison between the performance of the proposed algorithm and of the strengthened MIP formulation. Findings are that the proposed approach is better scalable. Additionally, it is competitive in terms of computing time. Further research will focus on testing the promising Lagrangian approach with larger problem instances as well as on combining the algorithm with a TS approach.

### *3.1.3 Crane split*

Conventional quay cranes have a lift capacity of one container. Operations of (un)loading a vessel and (un)loading a vehicle, such as an AGV, can be decoupled using double trolley cranes with a particular transit platform serving as interface or buffer for handshaking containers between the aft trolley and the fore trolley. Double trolley quay cranes operate, e.g., at the Container Terminal Altenwerder (CTA), Hamburg (Germany).

New developments are cranes with twin or tandem lift ability (see, e.g., Johansen (2006); Shanghai Zhenhua Port Machinery (2007b)). Four adjacent 20-ft containers

**(a)** Twin 40-foot quay crane operating on parallel trucks (courtesy of Shanghai Zhenhua Port Machinery – ZPMC, www.zpmc.com)

**(b)** Trailer holding four units of 20-ft containers at a time, port of Dubai (courtesy of ZPMC)


**(c)** Twin lift of containers (here: dual trolley crane; courtesy of ZPMC)


**(d)** Twin lift quay crane serving a tandem chassis (courtesy of ZPMC)

**Fig. 3** Twin lift handling

or two 40-ft containers, respectively, can be lifted at once. The cranes are equipped with a specific twin container spreader. Mechanical linkages between the two spreaders facilitate the adjustment to different container heights as well as to side-to-side clearances of two containers quayside while landing them onto adjacent yard chassis. Twin lift cranes are designed for faster (un)loading operations in order to meet the demands of mega-vessels. They are expected to boost the productivity by up to 50% if (un)loading techniques are improved accordingly. The twin handling can be supported by special trailer platforms or by trucks or AGVs operating side by side (see Fig. 3).

In Shanghai Zhenhua Port Machinery (2007c) the twin lift concept is combined with the dual trolley concept for considerable productivity increase. It is regarded as the most effective and fastest crane of the world for meeting the challenge of fast handling of current and upcoming mega-vessels. Theoretically, twin 40-ft double trolley container cranes are able to handle between 80 and 100 40-ft containers per hour. The aft trolley can serve one chassis/AGV or two chassis simultaneously. Twin 40-ft double trolley quay cranes are provided for the port of Qingdao (China).

**Table 1** Technical specifications of quay cranes (Ceres Paragon Terminal BV 2006; Dubini 2006; Konecranes VLC 2007 and Shanghai Zhenhua Port Machinery 2007a,b,c)

| | Conventional | Double trolley | Twin 40-ft single trolley | Twin 40-ft double trolley |
|---|---|---|---|---|
| Lifting capacity (t) | | | | |
| Double/twin spreader | – | – | 80 | 80 |
| Single spreader | 40–65 | 57–61 | 65 | 65 |
| Under cargo beam | 75–100 | | 100 | |
| Fore trolley speed (m/s) | 1.2–4.2 | 3.7–4.0 | 4.2 | 4.0 |
| Hoisting speed full | 0.4–1.5 | 1.2–1.3 | 1.5 | 1.5 |
| Hoisting speed empty | 0.8–3.0 | 3.0 | 3.0 | 3.0 |
| Aft trolley speed (m/s) | – | 4.0 | – | 4.0 |
| Hoisting speed full | – | 0.5–0.8 | – | 0.8 |
| Hoisting speed empty | – | 1.2–1.6 | – | 1.7 |
| Boom up time single (s) | 180–300 | | 300 | |
| Hoisting/lift height (m) | | | | |
| Fore trolley | | | | |
| Above rail top | 36 | 38.5–42.0 | 41 | ≥41 |
| Below rail top | 14 | 20.0–23.0 | 19.5 | |
| Aft trolley | – | | – | 15 |
| Maximum outreach (m) | 30–65 | 61–63 | 68 | ≥63 |
| Back reach (m) | 15 | 16–25 | 23 | 19 |
| Rail span (m) | 30.5 | 35.0 | 30.5 | 35.0 |

Technical specifications of quay cranes are presented in Table 1.

Kim and Park (2004) propose a branch-and-bound algorithm and a greedy randomized adaptive search procedure (GRASP) to solve the quay crane scheduling and load sequencing problem. The objective is the minimization of the weighted sum of the makespan of the container vessel and the total completion time of all quay cranes. It is assumed that berthing and departure times are given as well as the starting times for operations of the assigned quay cranes. Contrary to Peterkofsky and Daganzo (1990), Kim and Park separate the detailed crane scheduling problem from the berth scheduling problem in order to address the quay crane scheduling problem at a more detailed level by taking non-interference constraints into account. The problem is restricted to a single vessel. The computational complexity of the studied problem is not discussed.

Non-interference constraints are considered in early studies conducted by Lim et al. (2004b) and Zhu and Lim (2004) (see also Lim et al. (2004c,d, 2007) as well as Zhu and Lim (2006)). In Lim et al. (2004b) the problem is modeled as bipartite graph matching problem. Cranes and jobs are taken as vertices, and the weights of connecting edges are defined as crane-to-job throughput. A dynamic programming algorithm is proposed for solving problems with simple spatial constraints in order to find a static crane-to-job matching with maximum throughput. Furthermore, the problem is expanded with more complex constraints. This $\mathcal{NP}$-complete problem is solved by a probabilistic TS and a squeaky wheel optimization with local search. Since the model is based upon a "profit value" of a job which is difficult to define in practice, the implementation of the study in real-world terminals seems to be difficult. Furthermore, precedence constraints between tasks are not considered. The study is augmented in Zhu and Lim (2006) with regard to real-world practice aiming at completion of all the jobs with respect to certain criteria instead of only maximizing the total throughput without taking time into consideration. Thus, the objective is to minimize the latest completion time of all the non-preemptive jobs, which come in different sizes.

Li et al. (2006) provide an MILP for the quay crane scheduling problem for a mid-range planning horizon of one week. The objective is the minimization of the maximum relative tardiness of vessel departures. Due to its large size the model is difficult to solve directly. Therefore, a heuristic decomposition is proposed resulting in a vessel level model and a berth level model. The vessel level model provides the optimal processing time for any given number of quay cranes assigned to each individual vessel. The berth level model considers the entire set of vessels. Quay cranes are assigned among the vessels using the results from the given berth allocation and from the vessel level model. Two heuristics are discussed for solving different vessel level models with the same berth level model. Computational experiments on generated problem instances of different size are carried out for the evaluation of the heuristic approach against a lower bound. The results show the efficiency and effectiveness of the proposed approach. A finding is that insufficient quay crane resources are the main reason for long delays. Additionally, the tardiness of vessels can be reduced by allowing the splitting of a bay's workload. Further research will focus on speeding up the solution process as well as on investigating the interaction of the berth allocation problem and the quay crane scheduling problem. This approach will aim at obtaining a better solution for the integrated terminal resource scheduling problem.

Ng and Mak (2006) propose a heuristic for solving the problem of identical quay cranes moving on a linear rail. The approach aims at minimizing the ship's stay time in port by finding an appropriate work schedule of each quay crane. Only one crane can operate at a ship's bay at the same time. The problem is formulated as an integer programming model. It is decomposed by partitioning the ship into non-overlapping zones. The resulting subproblems for each possible partition can be optimally solved by a simple rule. Tight lower bounds for the minimum makespan are found by enhancing a lower bounding algorithm proposed by Webster (1996) for the problem of scheduling jobs with sequence independent processing times on identical parallel machines. Typical terminal operations data are used to generate a set of test problems for evaluating the heuristic algorithm's performance. The heuristic solutions are on average 4.8% above their lower bounds, with a mean deviation ranging from 2.1 to 7.3%. The maximum deviation ranges from 6.2 to 16.0%, the minimum deviation ranges from 0 to 3.7%.

Moccia et al. (2006) mention the difference between the time precedence constraint of the quay crane scheduling problem and the route precedence constraint in the pickup and delivery problem or the dial-a-ride problem. They formulate the quay crane problem as VRP with side constraints including precedence relationships between vertices. The problem formulation strengthens the model of Kim and Park (2004). The objective is to minimize the completion time of a vessel and the idle times of cranes caused by crane interferences. Large problem instances can be solved with the proposed branch-and-cut algorithm. Several families of valid inequalities are inserted for taking advantage of the precedence constraints between vertices. The paper indicates that the developed algorithm outperforms the algorithm proposed in Kim and Park (2004).

Lee et al. (2006a) study the $\mathcal{NP}$-complete quay crane scheduling with consideration of non-interference constraints. The study is stimulated from Kim and Park (2004). The objective is the minimization of the makespan of handling one single container vessel (i.e., the latest completion time among all holds). The developed GA obtains near optimal solutions for the proposed MIP model. It shows efficient performance in computational experiments on random instances of both small and large size.

Canonaco et al. (2007) present a queueing network model and a manager-friendly simulation tool for the management of container discharge and loading at any given berthing point. The approach aims at matching the maximization of productivity of expensive resources, such as quay cranes, with the minimization of a vessel's berthing time with an adequate rate of service completion. Experiments based upon an event graph-based stochastic discrete event simulation allow for an evaluation of outcomes of different policies regarding crane assignment and scheduling. The data used in the paper stem from a terminal at Gioia Tauro. The implemented framework supports easy performance of what-if-experiments for optimum seeking.

Liang and Mi (2007) propose a multi-objective model for the quay crane scheduling in the berth allocation planning problem. Solutions are obtained in numerical experiments with a multi-objective hybrid GA incorporating a priority-based encoding. The mathematical model strives to improve standard models known from literature that are criticized for their unrealistic assumptions. In particular, a fixed handling time being irrelative to the berth location is assumed. Furthermore, the workload balance of quay cranes is neglected even though the number of quay cranes do affect the handling time of a berth. Consequently, the proposed model considers the relationship between the handling time, the berth location, and the departure delay of vessels, with a fixed number of quay cranes at a berth and identical working time of each quay crane at the same berth. The first objective is the minimization of the service and delay times of vessels. The second objective is the minimization of the standard deviation of the quay cranes' working time. The multistage GA is based upon an adaptive weight approach. This approach guides the search pressure towards an ideal point in search space by readjusting weights with utilizing information from the population. The problem is divided into a first phase for serviced sequencing of a vessel, a second phase for berth selection, and a third phase for assigning different numbers of quay cranes to berths. The case study for numerical experiments shows three examples with four berths and seven quay cranes. It is based upon data of a terminal in Shanghai (China). Some conclusions are derived from Pareto solutions. The first experiment shows that the efficiency of a terminal system is affected by the number of quay cranes. With an increasing number of quay cranes assigned to a berth the number of vessels waiting to be serviced at that fast berth increases. The total handling time of a berth with more quay cranes is shorter compared to a berth with less quay cranes. This results in workload imbalances. Since the second objective aims at workload balancing the algorithm randomly assigns vessels to berths. The trade-off between the operation time, the waiting time, the delay time, and workload balance can be observed in the experiment. Further research will enlarge the problem instances and focus on dynamic problems in integrated terminal systems.

Linn et al. (2007) present an approach for predicting the quay crane rate using the artificial neural network paradigm of a multilayer perceptron with a backpropagation learning algorithm. The quay crane rate is chosen as the performance indicator for terminal operations. It is noted that accurate predictions support efficient execution of operations. Furthermore, the proposed models may be used to suggest remedial actions if terminal operations are affected by security measures or incidents. The paper reports results from a study conducted in a terminal in Hong Kong. Rule-based procedures and variants of linear regression fail due to complex non-linear relationships among input variables that are difficult to describe by simple functions. Three categories of factors affecting the quay crane rate are distinguished: the estimated workload in the predicted period, resources to be deployed in the predicted period (such as cranes and internal trucks), and the container yard status reflecting the general situation of container storage in the yard and workload distributions (such as, among other values, the container density, the weighted average distance to be traveled between berth and storage block). Different neural nets are developed for different types of vessels in order to obtain accurate results. The overall model has 18 inputs, the model for a specific vessel type has 24 inputs. Half year's data from the Hong Kong terminal is used for training (70% of data sets), test (24%) and validation (6%) of the models. Experiments show that the models tailored to specific vessel types provide more accurate results than the model for the overall quay crane rate. The accuracy is measured in terms of the relative error between the predicted rate and the true rate and the average relative error with respect to the number of records in the evaluation data set. Over-prediction and under-prediction is treated equally despite economic relevance.

Some new ports have begun to use a dual (or double) cycle strategy for operations of quay cranes based on experience. The concept of double cycling has been recognized in the industry for at least 10 years. However, determining the extent to which it has been used or tried is difficult to judge due to the lack of publicly available information. Only few scientific studies on dual cycling of cranes are published. Studies conducted by Goodchild (2005, 2006a,b) and Goodchild and Daganzo (2004, 2005, 2006, 2007) are the first ones. They provide upper and lower bounds of the dual cycle model. Compared to single cycle mode the dual cycle operation doubles the number of quay crane tasks in a cycle by allowing to carry a container while moving from the apron to the ship (loading move) as well as from the ship to the apron (unloading move). Thus, empty moves of the single cycle mode are used for carrying a container as well. The problem is formulated as a two-machine flow shop scheduling problem assuming that one quay crane consists of two separate cranes, one for loading, the other for unloading. This strategy aims at minimizing a ship's turnaround time by the improvement of the cranes' efficiency, thus increasing productivity and throughput. Even with constraints due to hatch covers the strategy can lead to a gain in productivity and a higher utilization of the expensive quay cranes tending to be the bottleneck of a terminal. This strategy for increasing capacity provides benefits at low costs in contrast to expensive activities such as renovating and adding terminals, constructing and expanding intermodal facilities, installing dual hoist cranes, and implementing new information technology (IT) infrastructure. For the dual cycle method neither additional infrastructure nor technology is required. Another advantage is that it can be quickly implemented and complements existing methods.

Zhang and Kim (2007) note that only the upper and lower bounds of the dual cycle model are discussed by Goodchild and Daganzo. Furthermore, they note that the suggested scheduling only takes stacks under a single hatch cover into account. One may question whether the statement of the general dual cycling model in the referenced studies (Goodchild 2005; Goodchild and Daganzo 2006) is appropriately reflecting real world situations and whether the suggested strategies are easy to apply in real situations. Consequently, the problem is reformulated as MIP model with additional focus on twin lift activities, i.e., the handling of two 20-ft containers at the same time instead of one 40-ft container. The general problem is decomposed in order to account for real situations regarding to hatch covers. The approach differentiates between an inter-stage sequencing (hatch sequencing) and an intra-stage sequencing, i.e., the sequencing of crane tasks in the same hatch. A heuristic is proposed and numerical experiments are conducted for five real cases from Busan for showing the effectiveness of the approach.

## 3.2 Storage and stacking logistics

Saanen and Dekker (2006a,b) investigate different stacking strategies by means of simulation. They focus on (transshipment) terminals with RTGs in combination with terminal trucks since these systems are regarded as being most used in the world. The impact of yard operating rules is evaluated. Furthermore, the authors strive to define a set of rules that will increase the stacking capacity neither with an increase in costs per move nor with a performance decline. The rules forming a multi-factor decision making process are based upon principles and constraints within an RTG terminal as well as on ideas from operating an RMG. The main simulation results are the waterside productivity level in moves per hour, the landside service time of trucks on the interchange points, the productivity of transport vehicles and RTGs in moves per hour, and the truck handling time at the stack from arrival until ready to depart. The results demonstrate, e.g., the impact of an increasing occupancy rate of the stack on the quay crane performance. Furthermore, the relation between the number of shuffles per outbound move and quay crane productivity, the relation between RTG gantry time per move and quay crane productivity, and the relation between yard density and RTG gantry time per move, are presented. One finding is that the difference between a refined traditional strategy and a simple random stacking strategy in terms of quay crane productivity is small (0.7 container lifts per hour over six weeks of operation). Interesting relationships are the negative correlation between the yard density and the productivity caused by increased RTG movements. The impact of the density on the gantry movements of RTGs is lower for the traditional stacking strategy doing consolidation of containers. But the overall impact of gantrying on productivity is higher because the terminal is more sensitive to higher yard density with regard to RTG gantrying. Thus, a refinement of the random strategy should focus on reducing the gantry movements by trying to consolidate as much as possible but without reserving any space. This may be realized by assigning bonus points to storage locations being close to storage locations where similar containers are already stored. The RTG is expected to move less since similar containers are usually loaded at the same time. It should be considered that large contingents of containers may require more than one RTG. Thus, consolidation at one location may result in lower productivity due to RTGs that are hindering each other. Future research will focus on refining this assignment considering both effects.

Dekker et al. (2006) explore different stacking policies for containers in automated terminals by means of simulation. A comprehensive overview of stacking policies used in practice is provided. Several scenarios with variants of category stacking where containers can be exchanged during the loading process are considered. Categories are defined by the export modality and the place of a container in a vessel. They are based upon the weight class, the destination, and the type of container. It is assumed for certain categories that containers can be exchanged during the loading process. The categories support stacking optimization and online optimization of stowage. Data from Europe Container Terminals (ECT, Rotterdam) lead to 45 different categories for jumbo ships (8,000 TEU) and 90 categories for deep-sea ships. Workload variants for ASCs are considered as well. Performance criteria for stacking policies are reshuffles and reshuffle occasions (undesirable), the number of situations with no positions being available (undesirable), and the workload of the ASCs determined every 15 mins. Overall results of numerical experiments allow for the comparison of several scenarios. They indicate that busy crane quarters are the most important performance measure. Furthermore, the percentage or reshuffles can be significantly reduced. It is concluded that the capacity of a set of ASCs (27 to 29 in total) is exceeded for jumbo ships. The stack configuration is not able to absorb the high quay crane productivity. Thus, many short-term bottlenecks occur. Another finding is that category stacking results in much better performance than random stacking. Furthermore, peaks in ASC workloads can be reduced by implementing a workload control variable and by stacking on piles close to transfer points. The stacking performance can be extremely improved by detailed simulation experiments of stacking operations. They are considered to be essential for the design of automated container terminals.

Hirashima et al. (2006) propose a Q-Learning algorithm for determining desirable export container movements in the buffer area for the reduction of a ship's turnaround time. The loading sequence of a vessel has to be taken into account. Therefore, the initial random positions of containers stacked in the yard have to be rearranged into a desired arrangement. The proposed algorithm is based upon the number of container movements for the material handling in the yard. The method provides several desired container positions leading to an improvement of the learning performance. Numerical experiments show that the algorithm outperforms conventional single-goal methods being currently in use in both small and large problem instances. The number of container movements is much smaller than in manually designed plans.

Kang et al. (2006b,c) propose an SA-based approach in order to find a good stacking strategy for export containers with uncertain weight information. Computational experiments show that the new strategies outperform the traditional "same-weight-group-stacking" strategy since the number of time consuming rehandling operations is significantly reduced. Additionally, it is shown by numerical experiments that the application of machine learning algorithms for building good classifiers can result in a more accurate weight estimation leading to even better solutions. Future research will focus on a cost sensitive learning for the weight classification.

Kang et al. (2006a) propose an SA approach for solving the task of rearranging export containers within a block in the yard (remarshalling) with multiple non-crossing yard cranes. A remarshalling schedule free of rehandling operations during the loading of a vessel and during the remarshalling is based upon a partial order graph. The SA minimizes the time required for remarshalling at the level of that graph. Furthermore, an evaluation heuristic with a depth-limited branch-and-bound search for the construction of a crane schedule is proposed. Numerical experiments with a block of 33 bays, 9 rows, and 6 tiers, and 2 non-crossing cranes show that the proposed method is able to generate an efficient schedule in a reasonable time. Further research will focus on generating neighboring containers to be swapped taking feedback information of the evaluation heuristic into account.

Kim and Hong (2006) propose two methods for determining the location of relocated blocks in block stacking systems, e.g., containers in a container yard. The objective is the minimization of the number of relocations during the pickup operation of all containers in a yard-bay. Based upon a given initial configuration of a yard-bay and given pickup priorities among containers, storage locations of relocated containers as well as the pickup sequence of containers are determined. It is assumed that the precedence of pickups among containers is known, and that containers are relocated to other stacks in the same bay. Furthermore, pre-relocations are excluded. Two cases are distinguished: either precedence relationships among individual containers or among groups of containers are given. A branch-and-bound algorithm is provided. Since it may be inappropriate for practical real-time application it is compared with a heuristic rule. The rule is based upon an estimation of the expected number of additional relocations of a container considering the expected future relocations of containers from other stacks in the same bay to empty slots of the corresponding stack. Experimental results with 40 randomly generated instances for each of 13 problems with different sizes show that the heuristic rule outperforms the branch-and-bound approach in terms of average computation time. Furthermore, the computation time for the branch-and-bound algorithm increases while it is on the same level (less than 2 s) for the heuristic rule. The average total number of relocations is on average 7.3% higher for the heuristic rule compared to the branch-and-bound algorithm for the case of precedence relationships among individual containers, and 4.7% higher for the case of precedence relationships among groups of containers. Overall, the latter case shows a significant reduction of relocations. Furthermore, it is shown that the number of relocations is more sensitive to the number of tiers than to the number of stacks.

Kim and Lee (2006) apply the constraint satisfaction problem technique to the problem of allocating storage space to export containers. The objective of the pre-assignment of space is the maximization of the equipment's efficiency in loading operations. The constraints are collected from a terminal at Busan. Numerical experiments are conducted for the evaluation of the performance of the developed algorithm and the comparison of various variable-ordering rules in terms of computational time. One finding is that sequencing space demand (requirement) units by the stage criteria first, the vessel criteria second, and the size criteria third, results in the shortest computational time. Furthermore, the computational time is significantly affected by the value-ordering rule as well as the sequence of constraint propagation during the search process.

Kozan and Preston (2006) propose a cycle-fashioned model of a seaport terminal system in order to simultaneously determine an optimal storage strategy and a corresponding container handling schedule. The need for integration of two problems is caused by the fact that optimizing storage locations is pointless with transfers being far from optimal solutions and vice versa. The proposed novel iterative search algorithm integrates a container transfer model with a container location model. These sub-models have dependent decision variables. A GA, a TS algorithm, and a hybrid algorithm combining GA and TS, are developed for solving the problem. They are able to handle large problem instances of real size. Overall, the nonincreasing GA (with the same number of generations within each iteration) outperforms the hybrid approach by finding best solutions for a wide range of infrastructure configurations. The results of experiments are analyzed and compared with current practice at the port of Brisbane, Australia.

Lee et al. (2006b) present an MIP model for storage yard management in transshipment hubs with intensive yard activity. The model's objective is the determination of the minimum number of yard cranes to be deployed and the determination of the storage location for unloaded containers. Therefore, reshuffles and traffic congestion are minimized. A sequential heuristic and a heuristic based upon column generation are developed for solving large-sized problem instances that cannot be exactly solved in reasonable time. A useful bound for quantifying the quality of suboptimal solutions is proposed as well. Numerical experiments show that the heuristics are able to find near optimal solutions. Future research may focus on the design of yard templates (which are assumed to be given in the current approach) and their integration into the yard storage allocation model. Furthermore, efficient methods for finding better or exact solutions may be explored, such as metaheuristics or different decomposition methods.

Holguín-Veras and Jara-Díaz (2006) extend their approach for determining space allocation and prices for priority systems in container yards (Holguín-Veras and Jara-Díaz 1999). The basic study incorporates an analysis of three different pricing rules (welfare maximization, welfare maximization subject to a breakeven constraint, i.e. second best pricing, as well as profit maximization). A continuous function with the storage amount and the yard's capacity as variables is used for taking the impact of congestions in a yard on the handling costs into account. A constant arrival of containers is assumed. The case with a capacity constraint and the number of arrivals being elastic to the price is considered in the extended study. The aim of the paper is to gain theoretical knowledge about optimal pricing at container terminals in order to enable terminal managers to examine the conceptual consistency of their implemented pricing schemes.

Aydın and Ünlüyurt (2007) investigate rehandling strategies in order to minimize the total time required to retrieve containers from a bay in the yard in a fixed sequence. A branch-and-bound approach is proposed that optimally solves the problem. Large randomly generated instances can be solved in reasonable time. Near optimal solutions are obtained by applying three alternative heuristics. Furthermore, cleaning moves, which are container movements not caused by the fact that a container beneath the relocated container is being retrieved, are discussed. They are promising for further reduction of the total retrieval time. Future research will focus on these cleaning moves, on container groups with different retrieval time as well as on algorithms capable of handling precedence among groups of containers.

Kim and Kim (2007) propose methods for determining prices for the storage of containers in a yard. A storage fee encourages customers to store containers only for a short time in the terminal's yard. Contrary to Holguín-Veras and Jara-Díaz (1999, 2006), the approach is based upon detailed cost models of handling activities for customers and terminal operators. Three different objectives are considered: (1) the maximization of the profit of the terminal operator without constraints, (2) the maximization of the operator's profit with a customer service constraint, i.e., the maximum level of the average waiting time of outside trucks must be satisfied, and (3) the minimization of the total public cost including the operator's as well as the customers' cost. It is assumed that the storage fee is proportional to the length of the storage time beyond a free-time limit. An optimal objective value is found with a free-time limit of zero. Numerical experiments are conducted based upon empirical data from Busan. Results show that the optimal no-free-time limit policy outperforms the policy in practice with a positive free-time limit with respect to the social cost minimization as well as the terminal's profit maximization. Further research may focus on better price schedule structures.

Lim and Xu (2006) propose a critical-shaking neighborhood search for the yard allocation problem. This $\mathcal{NP}$-hard resource allocation problem is introduced by Chen et al. (2002a, 2004) as extension of the berth allocation problem (Lim 1998, 1999). The objective is to minimize the yard space needed. The yard is treated as a one-dimensional space. Numerical experiments are conducted adopting benchmark instances provided by (Chen et al. 2002b) (available via http://www.comp.nus.edu.sg/~fuzh/YAP). The idea of the proposed heuristic is the iterative improvement of the priority sequence's quality from an initial random sequence. Some critical requests are picked and their priorities are randomly shaken. The shaking procedure is different from the squeaky wheel optimization which always increases the priorities of critical requests. After shaking the local search is explored. It is shown that the proposed heuristic is very effective. All benchmarks are improved or at least met in short time. The solution qualities are stable over different initializations. Furthermore, an effective greedy heuristic for generating initial sequences with high quality is proposed. Further research will focus on the application of the new approach on the $\mathcal{NP}$-hard general yard allocation problem.

The general yard allocation problem is proposed by Chen et al. (2003) (see also Fu et al. 2007) based upon the example of the port of Singapore. In this generic model, two-dimensional spaces are requested and have to be allocated within time intervals. The objective is the minimization of the total area used in the yard while satisfying all space requests. The problem is similar to the $\mathcal{NP}$-hard two-dimensional rectangular packing problem with a time dimension. A simple bottom left packing strategy is used as a first heuristic. Its results are improved by additional application of metaheuristics such as squeaky wheel optimization, TS, SA, and a GA. Numerical experiments on randomly generated data (available via http://www.comp.nus.edu.sg/~fuzh/GYAP) show that the best results are achieved by the GA. The second best approach is the SA, while the performance of the TS and the squeaky wheel implementations are poor.

Kim et al. (2007) focus on the container stacking policy in the yard of automated terminals. The proposed approach employs multi-criteria fuzzy decision-making methods with dynamic online adjustments of criteria weights reflecting the relative importance of the criteria. The current container yard configuration, the workload of automated transfer cranes (ATCs), the ATC interference probability, the expected stacking yard configuration, and future ATC operations utilizing information about stacked containers are taken into account. Thus, a container stacking strategy has to apply complex real-time processing. Different strategies regarding block determination and slot determination are considered. While the block can be assigned based on random, on current workload, or on distance to berth, the slot can be determined based on random as well, on the container category (e.g., size, weight, destination port, etc.), on the levels (striving at uniform stack heights), or on the distance to the transfer point. The performance is evaluated by means of simulation experiments with two non-crossing ATCs showing reductions of horizontal transport distances and waiting times of AGVs and external trucks. The multi-criteria approach outperforms a stacking policy with only a single criterion. Furthermore, the approach with dynamic weight adjustment outperforms stacking policies with uniform weights.

Lee and Hsu (2007) propose an integer programming model for the container pre-marshalling problem for a single ship and a yard served by a RMG. The solution of the problem determines a plan to rearrange export containers in the yard with consideration of a given yard layout and a given loading sequence so that no extra re-shuffling is needed during the loading operation. The objective is the minimization of the number of container movements during the pre-marshalling. The model is composed of a multi-commodity network flow model and a set of side constraints representing physical laws that containers have to comply. Within the network, the nodes and arcs correspond to the yard's time-space structure whereas the flows represent the containers' movements in time-space. Several extensions of the model and the heuristic are discussed. Computational results of experiments with 37 examples are provided. Future research may focus on an extended problem formulation with multiple ships.

## 3.3 Transport optimization

### *3.3.1 The quayside transport*

*Automated guided vehicles* AGVs are commonly used in warehouse operations and flexible manufacturing systems. Therefore, a wealth of references regarding AGVs can be found (see, e.g., Qiu et al. (2002) for a survey of scheduling and routing algorithms, or more recently Ho and Hsieh (2004); Ho and Chien (2006); Lee and Srisawat (2006)). Technical specifications of an average AGV are shown in Table 2.

Vis (2006b) presents a comprehensive survey of research focused on design and control of AGV systems. It is documented that research focuses on manufacturing systems. Analytical and simulation approaches are present in the literature. Several categories of methods such as mathematical programming, queueing theory, network models, or

**Table 2** Technical specifications of an average AGV (Gottwald Port Technology 2007)

| | |
|---|---|
| Load types: Containers (ft) | |
| Standard | 40 or 2 × 20 |
| Symmetrical or asymmetrical | 20 |
| As option | 30 and 45 |
| Loading weights (t) | |
| Single container | 40 |
| 2 × 20 ft | 60 |
| Dimensions (approximately) | |
| Length (m) | 14.8 |
| Width (m) | 3.0 |
| Loading area height (m) | 1.7 |
| Deadweight (t) | 25.0 |
| Speed (m/s) | |
| Maximum forward/reverse | 6.0 |
| Maximum in curves | 3.0 |
| Traveling in crab mode | 1.0 |

Markov decision making as well as heuristics are employed in the research. Furthermore, it is concluded that most papers address one or two decision problems simultaneously while hardly any paper considers the relationship between AGV systems and other material handling systems. Finally, future research directions are identified.

Furthermore, approaches such as immunity-based systems are applied to control AGVs (see, e.g., Lau et al. (2007) for a robust and flexible warehousing system). In analogy to the human immune system, an artificial immune system can be implemented as a self-organizing multi-agent distributed system having distributed memory as well as specific mechanisms for learning behaviors.

Kim and Bae (2004) and Bae and Kim (2000) propose an extended approach for AGV dispatching with a network-based MIP model developed in Kim and Bae (1999). The objective is the minimization of the total idle time of a quay crane resulting from late arrivals of AGVs as well as the associated total travel time. It is assumed that storage locations of containers and schedules for (un)loading operations by quay cranes are given. The extension considers a pooled dispatching strategy taking multiple quay cranes and dual-cycle operations of AGVs into account.

Grunow et al. (2004a,b) focus on dispatching multi-load AGVs. A flexible priority rule-based approach is proposed and compared to an alternative MIP formulation in different scenarios. It is shown that the AGVs' lateness can be reduced in case of using the multi-load mode. An improvement of the terminal's overall performance is expected. In addition, an MIP model is developed that allows determining optimal solutions for small problem instances. A hybrid approach using the MIP combined with fast heuristics on some special dispatching requests is suggested for real applications. A different MIP formulation can be found in Schneidereit et al. (2003). In Grunow et al. (2006) a scalable simulation model is used for the investigation and evaluation of dispatching strategies. The detailed simulation study takes stochastic handling times, various terminal configurations as well as operation modes of AGVs into account. The developed pattern-based offline heuristic with dual-load AGVs carrying two 20-ft containers instead of a single 40-ft container clearly outperforms conventional online heuristics adopted from flexible manufacturing systems. With dual load, a deviation from a lower bound of less than 5% is achieved for the most realistic scenario. The dual-load mode is employed for more than 30% of all 20-ft containers. Thus, the dual-load capability is essential for obtaining good results. Furthermore, the results of the numerical experiments indicate that all dispatching strategies are rather insensitive to the size of the terminal configuration. The requirements of real-time dispatching are met. Further research may focus on the improvement of heuristics (neighborhood search).

Bish et al. (2005) focus on the quayside process of discharging and uploading containers to and from a single vessel. They discuss the problem of dispatching single load vehicles to the containers for the minimization of the ship's total serving time, i.e., the total time for discharging and uploading all containers. It is assumed that a fleet of vehicles is already assigned to the vessel. Other aspects of the yard processes, such as the assignment of an appropriate location for an unloaded container, vehicle routing, or traffic control, etc., are treated as given inputs. Several variations of a greedy algorithm are proposed that can be easily implemented in particular for large problem instances. For characterizing the effectiveness, the paper is focused on simple vehicle dispatching rules. Analytical bounds on the deviation of the heuristic solution are derived from the optimal solution for any problem instance. The absolute and asymptotic worst-case performance ratios of the proposed heuristics are identified. Most of the proposed algorithms are optimal in simple settings with a single ship and a single quay crane. In more general settings with multiple quay cranes the results of numerical experiments on test problems with four vehicles and two cranes with a job list of 8–12 jobs are near optimal.

Möhring et al. (2005) propose a real-time algorithm for AGV routing. Collisions, deadlocks and livelocks are avoided at the time of route calculation. The approach is based upon the determination of a shortest path with time-windows for each request and a subsequent readjustment of the time-windows. Computation times for the conflict-free routing are appropriate for real-time applications. In terms of overall transit times the algorithm is superior to a static approach used at the CTA in Hamburg, in particular for scenarios with many AGVs causing heavy traffic.

Zhang et al. (2005) present three mixed 0–1 integer programming models for dispatching vehicles such as AGVs or yard trucks at the quayside. The models consider the unloading phase of a vessel in one berth without taking physical settings such as buffers for vehicles into account. Furthermore, a congestion free vehicle traffic is assumed, and continuous operations of quay cranes are not guaranteed since the number of vehicles is limited. The models determine the starting times of the unloading operations as well as the order of vehicles for carrying out the jobs. The objective is the minimization of the overall waiting time of the quay crane (or container jobs) which is equivalent to minimizing the job ready time of the last job. Numerical experiments with up to 5 vehicles and 50 container jobs show that the best solutions are obtained by the proposed greedy algorithm. It is capable of solving even large scale problems with up to 100,000 jobs quickly.

Briskorn and Hartmann (2006) and Briskorn et al. (2006) propose a dispatching strategy for AGVs with a real-time job-vehicle assignment. It is based on a formulation with rough analogy to inventory management. This is in contrast to the common formulation with due dates for jobs. The alternative formulation is motivated by avoiding the use of estimated times, such as driving times, completion times, due dates, and tardiness, to a large extent. In practice, those estimates hinder accurate and robust planning and require complex terminal control systems with frequent updates of times. Furthermore, the novel approach enforces dual cycling of AGVs at stacks that reduces empty travels and lead to productivity gain at the waterside. The proposed problem formulation can be solved exactly. A simulation study shows higher productivity at a terminal applying this strategy compared to conventional due-date-based strategies. The good results for AGV dispatching motivate further application to other container handling equipment, such as stacking cranes and SCs.

Current research on deadlock handling at automated container terminals can be found in Kim et al. (2006) and Lehmann et al. (2006). In the former paper, Kim et al. propose an efficient algorithm for deadlock prediction and prevention in AGV systems. The approach guarantees deadlock-free schedules for AGVs to cross the same area at the same time. The method is evaluated in a simulation study showing satisfactory results. Average speed of vehicles, space utilization, and computational time imply potential for the method to be used in practice. Lehmann et al. (2006) address blocking effects between vehicles and handling units since common literature focuses on deadlocks only with regard to routing of AGVs and their guide path. A comprehensive simulation study is conducted. It shows the suitability of different methods for the detection and resolution of deadlocks occurring in the phase of resource assignment. The proposed deadlock-handling scheme is regarded as "a first step towards integrated scheduling and dispatching approaches for equipment units in highly automated container terminals." The approach with resolving deadlocks rather than entirely avoiding them is preferred as being "the most appropriate alternative, as more conservative approaches would result in lower equipment utilization." It is pointed out that the implementation of the proposed methods into the logistics control software of automated terminals is necessary (if not economically viable) in order to avoid a downtime of an entire terminal.

Duinkerken et al. (2006b) compare different trajectory planning strategies for AGVs by means of simulation. Experiments show a high potential for a dynamic evasive free ranging approach (in contrast to approaches with fixed routes) that improves a system's transport capacity and accuracy. The use of the full potential of the AGVs' free ranging capacities results in benefits of the dynamic approach. On the strategic level the discrete event simulation model compares this approach using shortest connections and a two-dimensional collision detection algorithm with several common fixed layouts. Further research will concentrate on a deeper investigation of cross-over variants as well as on collision avoidance for mesh and loop variants. Furthermore, tactical and operational controllers will be developed. Finally, the influence of stochastic processing times on the overall terminal performance will be studied. This stochasticity occurs, e.g., for crane handling, navigation systems, and communication delays.

**Table 3** Technical specifications and costs of an average SC (Ceres Paragon Terminal BV 2006; Noell Mobile Systems and Cranes 2007a,b; Siemens 2007c and Vis 2006a )

| | |
|---|---|
| Driving speed (m/s) | 5.5–8.0 |
| Hoisting speed full (m/s) | 0.3–0.4 |
| Hoisting speed empty (m/s) | 0.4–0.5 |
| Maximum stacking height (boxes) | 4 |
| Span width (boxes) | 1 |
| Position to change rows | head/tail of row |
| Distance required to change rows (m) | ≈ 14 |
| Maximum lifting capacity (t) | 40–60 |
| Purchasing costs US$ | ≈ 600,000 |
| Driver US$/h | 35–40 |

*Straddle carriers* SCs are engaged in different types of complicated container handling. Therefore, their efficient routing is achieved by minimization of empty runs. Only a few papers address the routing problem regarding SCs (Steenken et al. 2004). Technical specifications and costs of an average SC are shown in Table 3.

Das and Spasovic (2003) present a scheduling procedure for SCs. An assignment algorithm dynamically matches SCs and trucks as they become available. The objective is the minimization of empty travels and delays in servicing customers. The superiority of the proposed procedure over two alternative scheduling strategies is shown by tests with a simulation model of a real-world system (New York/New Jersey).

*Automated lifting vehicles* An automated lifting vehicle (ALV; also denoted as automated straddle carrier) has the superior handling capability of lifting a container from the ground. No additional lifting equipment is necessary. A buffer area is used as an interface to a quay crane (see, e.g., Vis and Harika 2004; Yang et al. 2004). However, an ALV system is employed only by the port of Brisbane so far. It has been introduced in December 2005.

Nguyen and Kim (2007) discuss the dispatching of ALVs. Information about locations and times of future delivery tasks is utilized in an MIP model aiming at an optimal assignment of delivery tasks to ALVs. The $\mathcal{NP}$-hard problem is formulated as a scheduling problem with precedence and buffer constraints similar to the multiple traveling salesmen problem with precedence constraints and time windows. The problem has three particular properties: (1) tasks must be performed exactly in coincidence with the planned sequence of loading and unloading operations; (2) the aim is the minimization of delays of quay cranes since the quay cranes are the expensive bottleneck resources. The minimization of the ALVs' total travel time is less important; (3) a delay of a quay crane transfer operation results in a delay of all succeeding transfer operations of the same crane by the same amount of time due to the predetermined sequence of operations. An ALV transfers one container at a time. It can be used in a pooling strategy, i.e., one ALV can serve multiple quay cranes. Since automated yard cranes as well as a buffer area usually are not bottleneck resources the queueing time of ALVs is negligible. It is not taken into account in order to avoid additional complexity. The transfer times in dual cycle operation are negligible as well. Additionally, a buffer area is not necessary for the handshaking between a quay crane and an ALV, and congestions on ALV guide paths are not considered. The proposed heuristic algorithm converts buffer constraints into time window constraints. Thus, an appropriate second heuristic is suggested for solving that converted problem. The approach is tested in experiments with generated problem instances differing in the number of ALVs (1, 2), number of quay cranes (1, 2), number of operations for each crane (4, 6, 8, 10, 12) and the buffer capacity (1, 2). The results are compared to the optimal results provided by CPLEX® showing an average factor of 1.01 (i.e., the heuristic's objective value is on average 1.01 times higher than the optimum) and a maximum factor of 1.086. Future studies with extended problem sizes will include the computational time for evaluation purposes as well as a dynamic environment and the synchronization of ALVs with cranes.

### *3.3.2 The landside transport*

While the basic VRP and lots of variants have attracted the attention of many researchers only a few papers are focused on problems with trucks and trailers (see, e.g., Chao 2002; Scheuerer 2006; Tan et al. 2006). Chao (2002) tests a TS method for improving solutions that are initially found by a construction method. Twenty-one instances of the truck and trailer routing problem are considered based upon seven basic VRPs from the well-known CMT-set of test problems (Christofides et al. 1979). Scheuerer (2006) introduces two simple efficient construction heuristics and a TS heuristic for solving the truck and trailer routing problem proposed in Chao (2002). While computational results show that the new heuristics are competitive to the existing approaches, the TS algorithm obtained better solutions for each of Chao's 21 benchmark problems. Tan et al. (2006) introduce a transportation problem for moving empty or laden containers for a logistic company with a limited number of trucks and trailers. A hybrid multi-objective evolutionary algorithm is applied to find the Pareto-optimal routing.

To the best of our knowledge the number of papers focusing on trucks and trailers at container terminals is very limited.

Nishimura et al. (2005) propose a trailer assignment method for solving the dynamic trailer routing problem at maritime container terminals where yard trailers are usually assigned to specific quay cranes until the work is completed. A new dynamic routing scheme is shown. It aims at saving yard operation time as well as container handling costs. The paper examines the problem of pickup and delivery with multiple tours being independent and not connected at a depot (like in the standard pickup and delivery problem). The trailer routing is defined with a given set of calling vessels. A new routing decision is made when a ship starts loading or discharging. A fleet of trailers has a set of tours connecting quay cranes and stack points in the yard. A static tour is a shuttle transit between shipside and landside transfer location. For example, a single-trailer picks up a container at a discharging quay crane and delivers it to an assigned stack area for unloaded containers. Finally it returns to the quay crane. A trailer's usage is much more flexible in a dynamic tour with moving the trailer to a different unloading quay crane than before or to a stack area for export containers in order to transport a container to a loading quay crane. The most flexible itineraries with mixtures of pickup and delivery can be achieved with multi-trailers in dynamic tours. Computational experiments on test data show that the dynamic trailer assignment is superior to the static version in terms of capital and operating terminal costs. The fleet size can be reduced due to shorter empty travel distance of trailers. The dynamic assignment principle is suggested to be implemented for both tactical and operational decisions within terminal management. For planning new terminals, ship handling and trailer routing can be simulated in order to determine the trailer fleet size. The simulation of trailer movements can also be useful for stevedoring companies making up a daily/weekly trailer work schedule given a prospective cargo handling profile. A disadvantage is the complexity of the itineraries so that errors of trailer drivers may increase. But these types of errors may be reduced by the application of modern communication and tracking systems.

Ng and Mak (2004) propose an algorithm for sequencing trucks that have to enter the working lane adjacent to a yard block with export containers. The objective of this approach is to reduce congestions of the working lane by minimizing the total time required to serve all empty trucks that are dispatched to a yard block.

Ng et al. (2007) address the problem of scheduling a fleet of trucks at a container terminal in order to minimize the makespan. The trucks have to perform a set of transportation jobs with sequence-dependent processing times and different ready times. The formulated $\mathcal{NP}$-hard MIP problem is solved by use of a GA. The GA's performance is enhanced by incorporating instance-specific information in the search process. In the truck scheduling problem, the travel time between two locations, the truck's ready time, the job's ready time as well as the duration of each job are taken into account. Useful information is inherited by a developed greedy crossover scheme aiming at the reduction of computational effort. This novel scheme is tested against six popular schemes (partially mapped crossover, order crossover, position-based crossover, order-based crossover, a fast union crossover called "union crossover #2" (introduced by Poon and Carter 1995), and enhanced edge crossover) which have been shown in other studies to be effective for solving parallel machine scheduling problems, VRPs, and traveling salesman problems. The performance of the crossover schemes is evaluated by solving a set of randomly generated test problems based upon real-life terminal operations data. Therefore, the test data reflect typical technical figures, e.g., a truck speed of 15 km/h, 2 mins handling time for a quay crane per container, 4 mins for a yard crane, etc. The new GA's solutions are on average 4.05 % better than the best solutions of the other six GAs.

### *3.3.3 Crane transport optimization*

The optimization of transports performed by gantry cranes operating in stacks focuses on sequencing of jobs and their assignment to the respective crane. Priorities of jobs have to be taken into account. Transport optimization for stacker cranes reduces to the same requirements as for the horizontal transport. Comparative algorithms can be applied. A common objective is the minimization of the waiting times of the transport vehicles at the stack or bay interfaces and the travel times of the cranes. Crane operations at the stack and operations at the quayside or landside are interdependent. Since traffic at the interfaces changes rapidly, online optimization is demanded for and job

**Table 4** Technical specifications and costs of an average RTG, RMG, DRMG, Twin RMG (Cederqvist 2006; Kalmar Industries 2007a; Saanen and Valkengoed 2005; Siemens 2007a,b and Vis 2006a)

| | RTG | RMG | DRMG | Twin RMG |
|---|---|---|---|---|
| Driving speed (m/s) | 2.0 | 4.0 | 3.0/3.5 | 4.0 |
| Trolley speed (m/s) | 1.2 | 0.8–1.1 | 1.0 | 1.0 |
| Hoisting speed full (m/s) | 0.3 | 0.3–0.6 | 1.0 | 1.0 |
| Hoisting speed empty (m/s) | 0.7 | 0.6–1.0 | 1.5 | 1.5 |
| Dead times between movements (s) | | 2 | 2 | 2 |
| Positioning time on a truck (s) | | 30 | 30 | 30 |
| Positioning time on an AGV (s) | | 10 | 10 | 10 |
| Stacking height (boxes) | | $\leqslant 4$ | $\leqslant 10$ | $\leqslant 10$ |
| Span width (boxes) | | $\leqslant 6$ | $\leqslant 10$ | $\leqslant 10$ |
| Maximum lifting capacity (t) | | 60 | | |
| Purchasing costs (Mio US$) | >1 | ≈2–2.5 | | |
| Driver (1,000 US$/man per year) | 25–100 | 0 | 0 | 0 |
| IT staff (US$/h) | | ≈115 | ≈115 | ≈115 |

sequences have to be recalculated whenever a new job arises. Technical specifications and costs of an average RTG, RMG, Double RMG (DRMG) and Twin RMG are presented in Table 4.

Only a few papers address the routing problem regarding gantry cranes at stacks within container terminals.

Ng and Mak (2005a,b) address the problem of scheduling a yard crane performing a given set of (un)loading jobs with different ready times. The objective is the minimization of the sum of job waiting times. A branch-and-bound algorithm is proposed for solving the $\mathcal{NP}$-complete problem. Numerical experiments with generated test instances based upon real-life data show that optimal sequences can be found for most instances of real size.

Most papers focus on operations of a single crane. Routing or scheduling algorithms for multiple cranes are hardly addressed in literature.

Saanen and Valkengoed (2005) compare three different configurations employing RMGs: the single RMG, the DRMG and the twin RMG (with two separated RMGs, one serving the waterside, the other serving the landside). Different stacking alternatives are evaluated by means of simulation in terms of throughput, flexibility, complexity, operational costs, and investment costs. Overall, the DRMG appears to be the best performing one but it needs the highest amount of space.

Ng (2005) studies the problem of scheduling multiple yard cranes in order to minimize the total loading time or the sum of truck waiting times in a yard zone, respectively. The problem is similar to the problem of scheduling DRMGs due to inter-crane interference with blocked cranes to be avoided. But it is not identical since DRMGs can pass each other on separated lanes whereas in Ng (2005) two or even more yard cranes share a single bi-directional traveling lane in a yard zone. The $\mathcal{NP}$-complete scheduling problem is modeled as an integer program. A heuristic based on dynamic programming and an algorithm to find lower bounds for benchmarking the heuristic's schedules are developed. Computational experiments show the effectiveness of the heuristic providing solutions on average 7.3% above their lower bounds. In Ng (2005) it is stated that "it is clear from the literature review that no studies have been conducted on scheduling of multiple yard cranes. Given the importance of yard crane operations on a terminal's productivity, effective yard crane schedules are needed." The approach does not consider loading sequence requirements.

Zyngiridis (2005) develops integer linear programs for scheduling one or two RMGs of equal size in a single block. The block is served by straddle carriers performing the (un)loading operations at the block's handshake areas. The optimal schedule satisfies all the demand on the landside and the waterside without delays. Furthermore, an optimal place to stack each import container is found. The primary objective is the satisfaction of all time commitments for export containers. Computational experiments based upon test data from Rotterdam are conducted for the evaluation of the performance of the RMG(s) working in blocks with different characteristics during a time period of 4h. Furthermore, the impacts of the block size and density of stacked containers in a block are tested. Findings are that the performance of one RMG is significantly affected by the length and the fill level of the block. The performance of two RMGs is only influenced by the length of the block. Besides, two RMGs always outperform one RMG which requires up to 70% more time than two RMGs. Further research will address the usage of other equipment as well as different block characteristics and stacking policies.

Cao et al. (2006b) discuss the scheduling of two yard cranes operating in two different yard blocks at the same time. They deliver containers to one quay crane performing loading operations. Therefore, loading sequence requirements are taken into account by simultaneous decision on visiting sequences as well as on the number of containers picked up at each visit of the two yard cranes. It is assumed that one yard crane only travels in one fixed block without transferring between blocks. A revised GA is tested in numerical experiments based upon generated sample instances. A comparison between its results and loosely estimated bounds shows satisfactory results. The dynamic deployment of yard cranes between blocks is identified for promising future research. A variation is proposed by Cao et al. (2006a) addressing a multiple yard crane system. A greedy heuristic and an algorithm based on SA are proposed to solve the $\mathcal{NP}$-complete problem of scheduling different yard cranes for minimizing the total handling time. The performance of the algorithms is tested through randomly generated small-scale and large-scale experiments. In small instances, 6–10 containers of two different types in a block of ten slots have to be picked up in three sub-tours. Large instances with 450–550 containers of five different types in a block of 45 slots to be handled in ten sub-tours are not solved exactly due to excessive computational time. Results show that the SA is outperformed by the greedy heuristic. Future research will focus on problem extension towards a system with multiple quay cranes.

Jung and Kim (2006) address the pickup scheduling problem of export containers for a terminal with multiple yard cranes and multiple quay cranes. It is assumed that the quay cranes' work schedules are given. The pickup scheduling determines the sequence of yard-bays for each yard crane to visit as well as the number of containers to be picked up at each yard-bay. The proposed loading schedule methods are based upon a GA and an SA approach considering interferences between adjacent yard cranes operating in the same block. The objective is the minimization of the makespan of the yard crane operation. The objective function is evaluated by considering the container handling time at a yard-bay, the yard crane travel time between two yard-bays (including setup time for positioning), and the waiting time due to yard crane interference. Numerical experiments result in the best set of search parameters of each algorithm. This best set is used in numerical experiments for comparison of the two heuristic algorithms. The results show that the SA outperforms the GA in terms of the average computational time and the average objective value. Further research may focus on the load sequence of individual containers within a specific yard-bay as well as on more detailed handling movements and constraints of yard cranes and quay cranes.

Jung et al. (2006) propose a GRASP algorithm for solving the quay crane scheduling problem. Congestions in the yard caused by interference of multiple yard cranes are taken into account. The objective is the minimization of the makespan of the quay crane operations. Computational experiments show that in particular the improvement phase of the heuristic search algorithm is too time consuming for practical application.

Lee et al. (2007) solve the problem of scheduling two yard cranes for loading export containers with the SA approach. The two gantry cranes serve the loading operations of one quay crane. They operate at two different container blocks by picking up a desired container from a block and loading it onto a yard truck waiting aside the block. The model aims at the minimization of the total loading time at the stack area. The schedule determines the container bay visiting sequences as well as the number of containers picked up simultaneously. Computational experiments show that the completion time found by the proposed algorithm is on average 10.03% above a loosely estimated lower bound. The performance of the algorithm is independent from the number of containers loaded.

## 3.4 Integrative approaches

Only a few studies regarding integrative views on container terminal logistics are published to date. They are based upon the awareness that improved terminal performance cannot necessarily be obtained by solving isolated problems but by an integration of various operations connected to each other. The studies can be subdivided into analytical approaches, simulation approaches, and approaches based upon distributed artificial intelligence or in particular multi-agent systems, respectively. Extensive integrative approaches may become more important with further increase of computational power. Currently, research is focused on quayside problems, such as quayside retrieval, and the closely related stowage planning.

### *3.4.1 Analytical approaches*

Murty et al. (2005) propose an approach with an integrative DSS. A variety of interrelated daily decisions at a container terminal is described and analyzed. The decisions aim at best use of storage space as well as at minimizing the vessels' berthing time, the resources, the external trucks' waiting time, and the congestions inside and outside the terminal. Ship turnaround time and quay crane throughput are regarded as key performance measures among other important measures, such as average waiting time of external trucks, the volume of unproductive moves, etc. Mathematical models and algorithms for solving problems and making satisfactory decisions are discussed, e.g., the optimal deployment of RTGs among the blocks or the optimal allocation of internal trucks to quay cranes. An online procedure for assigning storage positions to containers arriving for storage in a block is presented. Dynamic load attributes are incorporated into decisions on space allocation. The proposed DSS is evaluated at a terminal in Hong Kong showing a reduction of 30% in a vessel's turn-around time as well as a reduction of 35% in container handling costs.

Vis et al. (2005) investigate terminals with lifting vehicles and buffer areas with limited capacity. The objective is the minimization of the vehicle fleet size under time window constraints. Each container has a time window defined by a release time/due date in which the transportation should start. An integer linear programming model is developed. The analytical model's estimates of the vehicle fleet size are evaluated by means of a simulation model. Both results are similar, the analytical model performs well. It slightly underestimates the fleet size compared to the best fleet size obtained by the simulation approach ($\approx$ 10%). It is shown that these small underestimates have only small impact on the unloading times of the ship since the law of diminishing returns applies to the use of more vehicles. Unloading times are less sensitive to the fleet size if the fleet size approaches the optimal value.

Vidovic and Kim (2006) address three-stage material handling systems and the estimation of their productivity or cycle time, respectively. A container port terminal with quayside operations performed by quay cranes, yard trucks and yard cranes can be considered as a three-stage material handling system with no buffers between adjacent material handling equipment. A continuous Markov chain model is proposed as well as two approximated mathematical models based upon probability theory. Numerical experiments with real data from terminals in Busan and randomly generated data are performed for evaluating the accuracy of the analytical models. Findings are that the analytical approach provides better results than the Markov chain approach. Future research should be focused on the applicability to cases when the cycle times are not normally distributed. Furthermore, the multiple-handling-equipment problem with a few trailers working simultaneously in the second stage should be analyzed.

Alessandri et al. (2007b) propose a linear discrete-time model and a linear cost function in order to model the flows of containers through an intermodal container terminal and to optimize problems related to the strategic planning of maritime terminals. The objective of the proposed optimal control problem is the minimization of the transfer delays of containers in the terminal. The entire terminal is decomposed into three sub-terminals for ships, for trucks and trailers, and for block trains. Handshaking queues are used for describing delays in transferring containers from one resource to another one. A receding-horizon strategy is adopted in order to seek for a solution of the optimization problem. The effectiveness of the proposed control scheme is evaluated by numerical experiments using data from a Mediterranean port in the Northern part of Italy. Alessandri et al. (2007a) generalize the results of Alessandri et al. (2007b) by proposing a nonlinear predictive control approach for the allocation of the available handling resources in a maritime intermodal terminal. A mixed integer nonlinear problem is formulated for modelling the container flows within the terminal. Two solution techniques are developed. One approach treats decisions expressed by binary variables as non-differentiable functions. Thus, mathematical programming tools can be applied that do not need derivatives for the optimization. The other approach uses standard branch-and-bound algorithms for nonlinear programming. A simulation study with realistic data is conducted for the evaluation of the approaches with regard to computational effort, robustness with respect to local minima, as well as the effectiveness of the resulting predictive control in the equipment management. Results indicate that the first approach tends to trap in local minima. But the computational effort is very small. The branch-and-bound approach is limited by computational complexity in particular with long control horizon. Future research aims at improving the effectiveness of optimization.

Ak and Erera (2007) develop methods for the simultaneous assignment of berth space and scheduling of quay cranes taking the work distribution on vessels into account. The approach is based upon the papers of Daganzo (1989) and Park and Kim (2003). The objective is the minimization of the vessel's total service time and total lateness penalty. Small instances of the proposed MIP model with randomly generated test data are solved to optimality. Near optimal solutions for larger real size problems are obtained by a TS-based approach.

Chen et al. (2007a) propose a TS algorithm for the integrated scheduling problem of various kinds of container handling equipment at a maritime terminal. The problem is formulated as a hybrid flow shop scheduling problem with precedence and blocking constraints. The authors note that no study dealing with hybrid flow shops with unrelated parallel machines on each stage and sequence-dependent setup time as well as precedence and blocking constraints has been found. The objective is the improvement of the coordination among the equipment. Hence, the approach strives to increase the productivity of the terminal. The objective is expressed as the minimization of the makespan for serving a given set of vessels. For solving the problem optimally and having a benchmark for the evaluation of the heuristic algorithm's effectiveness, an MIP model and its lower bound are provided. A TS-based algorithm is developed for obtaining a better performance schedule. Computational experiments are conducted with generated small and medium-sized instances based upon real-life data from the port of Shanghai. One finding is that the TS with multiple insertion heuristics with a non-delay machine priority policy as initial solution generator outperforms the approach with a minimum extra time machine priority policy in terms of both the quality and the efficiency of the solution. Furthermore, results show the importance of a good initial solution heuristic for the scheduling problem.

Corry and Kozan (2006) discuss operating strategies for dynamic load planning of container trains at intermodal terminals. A load plan assigns containers to slots on a train. The authors strive to deal with typical uncertainties in the model parameters by means of numerical experiments on a simplified case as well as on a more realistic scenario in order to provide a foundation for future research. The paper is not focused on maritime terminals. Instead, a terminal where containers are transferred to/from trucks on a platform adjacent to transshipment tracks is considered. However, the proposed approach may help in gaining insight into operations at a maritime terminal's hinterland truck interface. The objectives of the load planning assignment model are the minimization of the handling time and the optimization of a train's mass distribution. The model is designed to operate within a rolling horizon due to the parameters' uncertainty.

### 3.4.2 Simulation approaches

Jin et al. (2004) present an "intelligent simulation method" based upon fuzzy artificial neural networks for the regulation of container yard operation including the system status evaluation as well as the operation rule and stack height regulation. The approach considers the operation scheduling by means of GAs combined with discrete event simulation. The first phase of the regulation process forecasts the quantity of incoming containers. The second inference phase decides on the operation rule and stack height. The operation scheduling is regarded as a fuzzy multi-objective programming problem with the objective of minimizing a ship's waiting time and the operation time. The usefulness of the method is shown in a case study with 2 berths, 64 blocks, a planning period of 24 h, and a prediction period of 3 days. A comparison between results of the proposed model and current operation in 30 days shows that the total ship waiting time is reduced from 64 h to 46 h.

Liu et al. (2004) investigate the impact of two commonly used terminal layouts and automation using AGVs on the terminal performance. One layout is characterized by container stacks that are placed in parallel with the berth. In the second layout, the stacks are arranged perpendicular to the berth. A multi attribute decision-making method is applied in order to evaluate the terminal performance and determine the optimal number of deployed AGVs. Three operational scenarios are considered and compared for both automated yard layouts: (1) loading, (2) unloading, and (3) combined loading and unloading operations. Simulation experiments based upon real-life yard operational data from Norfolk (USA) show that the performance of a non automated terminal can be substantially improved by automation using AGVs. An additional finding is that the yard layout influences the terminal performance as well as the number of AGVs. It is indicated that the combined operation can increase the terminal throughput as well as the utilization of equipment in the yard.

Veenstra et al. (2004) analyze economic aspects of a container terminal simulation. The simulation concepts help to show the interdependence of different decisions. The approach is useful for gaining insight into the decisions' influence on the overall performance of a terminal. Future research will focus on the integration of advanced operational and financial strategies, such as dynamic pricing, into the prototypic simulator.

Parola and Sciomachen (2005) propose a discrete event simulation approach for modeling the entire logistic chain of intermodal container flows in a port system network. The model is tested for a scenario in Liguria county (Italy). Particular attention is paid to the land transport and the modal split re-equilibrium. The approach aims at an evaluation of the present system configuration and the effects of a possible growth of container flows in a 2012 vision.

Bielli et al. (2006) propose an architecture and object-oriented model for the distributed simulation of container terminal operations in order to improve management practices and the efficiency of a terminal. The simulation software aims at providing a tool for the evaluation of (un)loading operations in terms of time and costs as well as of different storage strategies and resource allocation procedures. Therefore, operational indicators such as the global productivity, the net productivity, the utilization indices for quay crane, yard crane and shuttle, the yard occupancy rate, and the average ship waiting time are generated by the system. The simulator is calibrated and validated with data (2 weeks) of a terminal in Casablanca (Morocco).

Ng and Wong (2006) study the impact of vessel traffic interference in Hong Kong's terminal basin on the container terminals' capacities. They propose a simulation model taking navigation safety requirements (such as minimum clearances for a turning vessel) and terminal operational characteristics into account. One finding is that vessel traffic interference results in a considerable reduction of the terminals' capacities. The approach supports managerial decision making. It facilitates the estimation of the benefits of an improved movement coordination in the terminal basin, i.e., the inexpensive creation of additional terminal capacity. Capital intensive investments in infrastructure can be avoided.

Ottjes et al. (2006) focus on the design and evaluation of multi-terminal systems, such as Rotterdam's Maasvlakte port area. They propose a generic simulation model structure combining three basic terminal functions of transport, transfer, and stacking of containers. The modeling technique ("process interaction method") combines event scheduling and activity scanning for the identification of system elements and description of the sequence of actions for each element. The model is applied to existing and future terminals in Rotterdam. Experiments determine the requirements for the quay length, the storage capacity, and the handling and transport equipment for inter-terminal transport. The results of the study are used for further design of the terminals' infrastructure and for cost assessment.

Duinkerken et al. (2006a) propose a simulation model for the comparison of three systems with trucks and multi-trailers, AGVs, and ALVs for overland transport between container terminals within a large port area with several terminals, such as Rotterdam's Maasvlakte complex. The model incorporates a rule-based control system and an advanced planning algorithm. Numerical results, such as the utilization of vehicles or cost characteristics, for a realistic scenario for Rotterdam help in gaining insight into different characteristics of the transport systems and their particular interaction with the handling equipment.

Soriguera et al. (2006a) investigate the internal transport cycle in container terminal with SCs. The conflicts between the three subsystems of landside transportation, waterside transportation, and yard storage as well as related decisions are analyzed. A DSS for obtaining valid results for the entire transport chain is developed. A simulation approach based upon data from the port of Barcelona (Spain) is proposed. It enables the evaluation of single-cycle and double-cycle operation strategies for SCs as well as different fleet sizes of handling equipment. Numerical results are not provided.

Soriguera et al. (2006b) analyze the internal transport subsystem in a container terminal by means of queueing theory and simulation. The approach aims at gaining insight into the behavior of handling equipment as well as at optimizing operations. In particular, the focus is on the critical quayside. Experiments are based upon real data from the port of Barcelona. The results show that greater efficiency can be obtained by assigning the handling equipment fleet at the quayside to the berth as a whole instead of assigning it to an individual quay crane. But sophisticated operations planning and reliability are required in order to apply this assignment approach. Moreover, the impact of the human factor is pointed out. A decrease in productivity can be observed if a working area exceeds a single berth. Consequently, the cranes of a single berth should be regarded as an optimum cluster of cranes. The same equipment units should be assigned to each cluster for achieving the highest efficiency. Furthermore, the results indicate an estimated potential of cost savings of 3 US$ per TEU traffic on the terminal by applying the proposed guidelines for operating the internal transport subsystem. Future research will focus on including marginal costs of the stacking. This will facilitate a DSS on the strategic level since the best type of equipment for a predefined terminal layout can be selected.

Tu and Chang (2006) analyze operations at a ditch container wharf and container yards in future mega-container terminals by means of simulation. It is assumed that mega vessels with a capacity of more than 10,000 TEU will berth at only a few hubs in the world. Those hubs require fast entry and exit procedures for containers. Therefore, a ditch wharf for 15,000 TEU vessels is simulated in the three primary operation subsystems for ship-shore handling operations, storage operations in the yard, and the gate with reference to a terminal at Kaohsiung (Taiwan). From the results it is concluded that "in the operation of container terminals, the arrangement of practical movement lines in the entire area affecting the number of ship-shore cranes on each side and operation time for unloading containers, not a set allocation of time, will be more uniform." The time for container passing routes as well as time and orders of each operation machine should be considered for a more detailed simulation of internal subsystems.

Ha et al. (2007) present a simulation model for container terminals that reproduces detailed behavior of equipment. The approach is not restricted to movements of trucks and cranes. Movements of trolleys, spreaders, etc., are modeled and visualized in three-dimensional form with interactive capability as well. Different settings for equipment behavior, e.g., speed of trolleys and spreaders, can be evaluated with regard to the terminal's overall performance. Thus, the approach strives to support decisions on investment in new equipment by assessment of its effects on gaining the terminal's productivity subject to various constraints including limited budget and space. The proposed results are based upon a model with three quay cranes serving one berth with a single mooring vessel having a capacity of 9 bays, 10 rows, and 6 tiers. Transporters, such as AGVs, move containers between the berth and a storage yard according to predefined tracks. One container block with the dimensions of 20 bays, 6 rows, and 4 tiers, is served by a single yard crane. At the end of a block are transfer points, e.g., for external trucks arriving at the landside with exponentially distributed inter-arrival time. Thus, the assumptions reflect a common terminal layout. The first experiment investigates the effects of a varying speed of yard cranes and the number of yard tractors (with a cycle time for quay cranes fixed at 70.86, five blocks in the yard, and a tractor speed fixed at 3.8 m/s). The berth productivity cannot be improved with a yard crane speed higher than 5.5 m/s and about nine tractors. In this case additional investment in berth-side equipment is needed for further increase of productivity. Moreover, the experiment shows that the two factors are both bottlenecks since up to the observed stagnant point both factors can independently contribute to the productivity. A second experiment analyzes the effect of the number of blocks. Variations of four and six blocks show no substantial improvement in productivity.

A third experiment varies the number of blocks as well as the cycle time of the quay cranes (with the speed of yard cranes fixed at 2.5 m/s, and the number of yard tractors fixed at 10). Results show the impact of the number of blocks. It is noted but not surprising that the effect of the cycle time is negligible with a non-sufficient number of blocks while the increased yard crane performance results in a higher productivity when the yard-side capacity is sufficient.

### *3.4.3 Multi-agent approaches*

Thurston and Hu (2002) propose an agent architecture focusing on the quayside operations in order to reduce the ship handling time. The approach aims at a continuous improvement of an intelligent planning algorithm by dynamic and cooperative rescheduling of quay cranes and container vehicles. Although the paper is focused on the loading process the approach can be applied to unloading operations as well. The resources of the terminal are represented by three different types of agents for quay cranes, SCs, and traffic, and a fourth type for the superordinated area management. Future research will focus on the implementation of AGV speed control in order to realize advanced path reservation mechanisms. Based upon that implementation results from simulations using a constraint satisfaction approach will be compared with results of a greedy reservation algorithm intended to represent the commonly used first-come-first-served strategies.

A multi-agent based simulation approach for the evaluation of container terminal management operations is considered in studies of Henesey et al. (2002, 2003, 2006) and Henesey and Törnquist (2002) (see also Henesey 2004, 2006). The approach aims at planning and coordinating the processes within the terminal by mapping the terminal's objects and resources. The agents strive to complete their specified goal by searching, coordinating, communicating, and negotiating with other agents by means of a market based mechanism such as a series of auctions. In Henesey et al. (2003), several negotiating agents are proposed. For an arriving ship, a ship agent negotiates with berth agents in order to find the cheapest berthing position. Furthermore, the ship agent negotiates with yard agents representing different blocks and decides on purchase of appropriate storage space as well as on selling the discharged containers. The yard agents negotiate contracts regarding storage space and containers with the gate agent. Furthermore, quay crane agents, yard crane agents, and transport agents for transport vehicles are involved in the container handling process. The paper demonstrates initial ideas and concepts. Market-mechanisms or evaluation criteria are not discussed. In Henesey et al. (2006), experiments applying multi-agent systems for investigating the impact of different policies for sequencing, berthing, and stacking on the performance of container terminals are proposed. Numerical experiments based upon real data are conducted to evaluate eight transshipment policies. The finding is that a shorter vessel turnaround time can be achieved with good decisions on yard stacking (e.g., using the "stacking-by-destination" policy) and berth allocation. For some scenarios, the results show that some policies dominate other policies in terms of both ship turnaround time and shorter distances traveled by SCs. Future research will strive to integrate AGV coordination into the system as well as to optimize the berth allocation. Furthermore, economic or cost indicators may be incorporated in the system in order to support decisions based upon economic performance measures. For instance, the costs per hours for workers or working groups, the costs for fuel consumed by SCs, the number of containers handled during a particular period as well as the profit or loss made by certain operations may deliver useful additional information.

Franz et al. (2007) propose an integrative cost estimation concept and a multi-agent system approach for managing container terminals. In the current phase of the research the approach is applied to a limited problem scenario considering quayside transport vehicle scheduling and storage block allocation only. The holistic objective is the minimization of the average terminal-effected costs of container handling. The paper presents different market mechanisms for resource allocation by coordinating the market with bilateral polypolies. They are simple, flexible, efficient, and terminating. Thus, they meet the specific domain requirements such as short allocation times, the necessity of resource allocation, the dynamic operation schedules, the inherent distribution of information, the concurrency of activities of the market participants, the uncertainty about future demands and offers as well as the interdependency of allocations. The market mechanisms are based upon general auction types and include the extended reverse auction, the Dutch reverse auction, the continuous combinatorial reverse auction, and the continuous double auction. Beside these control mechanisms, a modular and expandable simulation system for the evaluation of the market mechanisms is demonstrated. An implemented prototype shows the validity of the approach. The combination of market-based control and multi-agent simulation is considered as a promising integrative approach for handling the complex operations within an entire container terminal. Future research will focus on the development of a productive multi-agent simulator for effective terminal analysis. Furthermore, the market mechanisms should be improved. An additional main task is the subsequent integration of the remaining terminal management problems.

## 4 Conclusions and outlook

An increasing number of theoretically and practically oriented papers during the last decade indicates the importance of optimizing logistic operations at seaport container terminals. Terminals are important in global trade. It is essential for operators to reduce unproductive demurrage at the port and to offer effective processes in order to meet the severe and increasing competition among terminals in this booming line of business with a high prospective growth rate. High investments as well as high operating costs for ships and port equipments enforce improvements of terminal operations. A terminal's competitiveness includes issues of waterside operations and internal logistics as well as landside operations, transport connection and routing within the surrounding area. A new challenge is the handling of mega vessels with a capacity of more than 10,000 TEU. Keys to efficiency seem to be the automation of in-yard transportation, storing and stacking as well as the application of optimization methods. For example, the application of intelligent routing and scheduling mechanisms for vehicles is part of this strategy. It allows for economic utilization of expensive equipment and space.

Research addresses more or less all elements of the transport chain within a terminal as well as outside a terminal. Strategic, tactical, and operational planning levels are considered. Modern information systems and communication technology enable the application of optimization methods in different areas of real terminals. However, the specific characteristics of a container terminal often hinder a direct application of models being abundant in standard literature and demand for model adjustments. Currently, most of the literature is focused on separated problems at a terminal, and mathematical models abstract from the entire transport chain and an intermodal network. Despite simplifications the models of restricted problems often remain complex. They can be helpful for gaining valuable insight and understanding of handling processes and problems within the entire system of a container terminal. While the need for holistic approaches and integrated optimization of operations in different terminal areas is identified, it is extremely difficult to solve real-world problems due to their complexity. The spectrum of methods successfully applied to a great variety of problems ranges from simple heuristics such as priority rule based approaches over metaheuristics (e.g., TS and GAs) to mathematical programming approaches and discrete event simulation.

Without doubt, technological progress in computers and communication systems supports the introduction and improvement of optimization-based DSSs. Restricted computing power constrains the solution and application of models reflecting important terminal characteristics by sophisticated algorithms. Furthermore, modern technology is necessary for the implementation of appropriate user interfaces for planners as a precondition for the acceptance of implemented methods. Although advances in information technology as well as the development of significant algorithms allow for finding good solutions for difficult problems in reasonable time, solving large integrated detailed models is beyond today’s computing capability. Therefore, decomposing the problem into several related smaller models is a common approach. Nevertheless, without any doubt, research has made considerable progress, e.g., in AGV handling, in quay crane operations by using dual cycling, or by considering multi-agent systems which are useful for the understanding and modeling of a terminal system.

Increased research on integrated optimization seems to be necessary for obtaining increased terminal performance. Without software incorporating optimization algorithms for control of terminal operations, novel equipment such as the Triple-RMG as well as cranes with twin-lift operation, or handling concepts for the container terminal of the future such as the floating crane (Anonymous 2006b), floating quay (Anonymous 2006a), the ship-in-a-slip concept with cranes simultaneously serving a vessel from two sides (Johansen 2006), or offshore container terminals will not necessarily lead to the desired and expected gain in productivity. Improved equipment must be complemented with “intelligent” software. The need for online optimization is another challenging topic.

Interesting components of an integrated approach may be methods receiving less consideration in container terminal oriented literature so far. For example, the problem of stochasticity is usually tackled by simulation. Additionally, stochastic optimization as well as scenario based planning may be fruitful research areas. Furthermore, the focus on extended variants of the vehicle routing problem could be a rewarding attempt (see, e.g., Hartl et al. 2006; Hasle 2003). Approaches such as vehicle routing with time windows and stochastic travel times or with stochastic customers (see, e.g., Bent et al. 2004; Wong and Leung 2002) may be useful and applicable at container terminals. Having test and benchmark data for simulation models as well as for optimization algorithms seems to be helpful for future research (see, e.g., Hartmann 2004).

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